

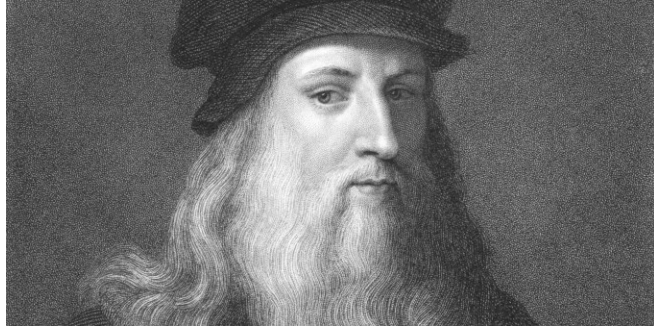
5G E2E QoS (Quality of Service)

Created: 2024_0518

Updated: 2025_0518

By JMA

Org_Resource

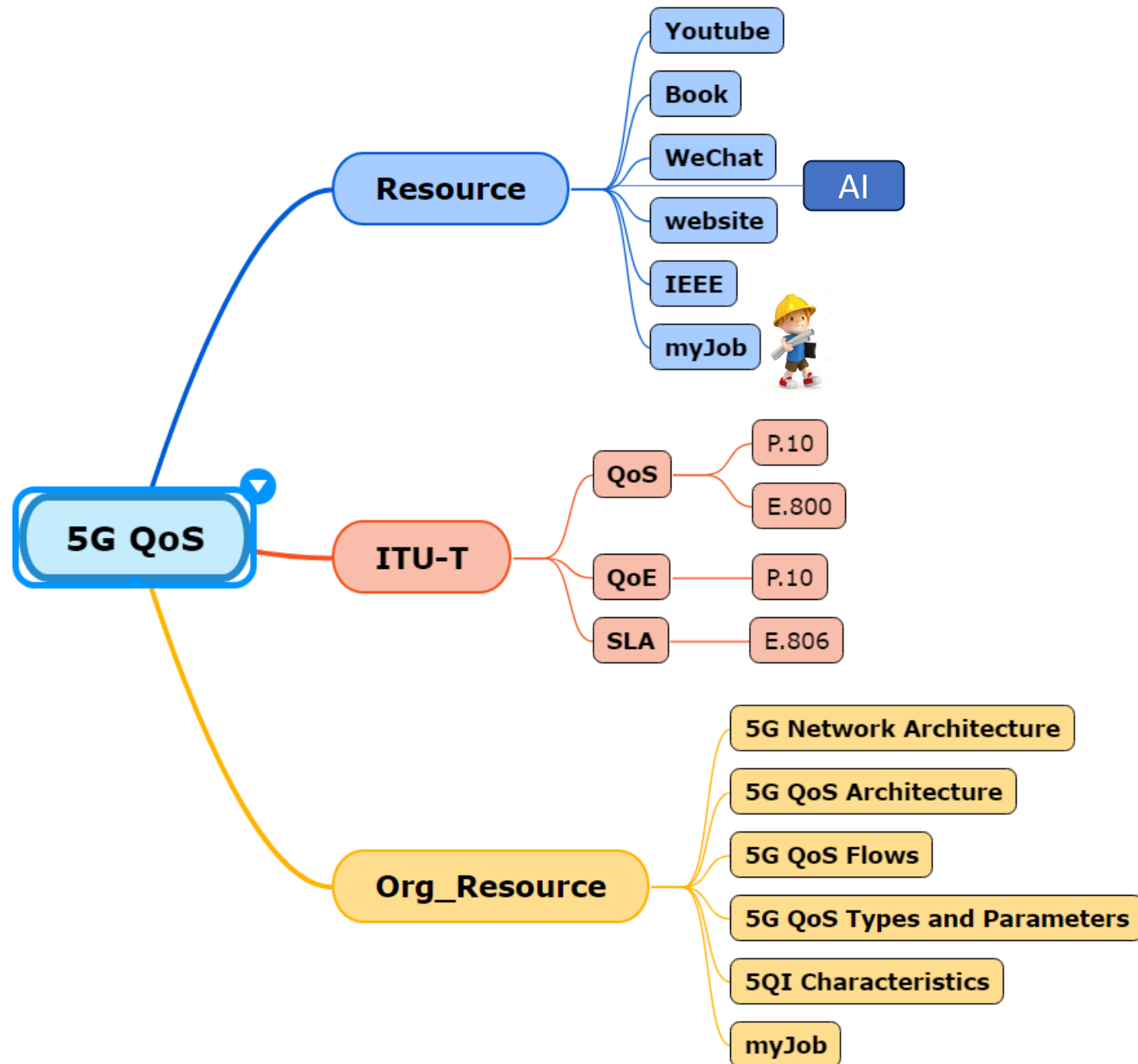


**He who has access to the fountain does not
go to the water-pot.**

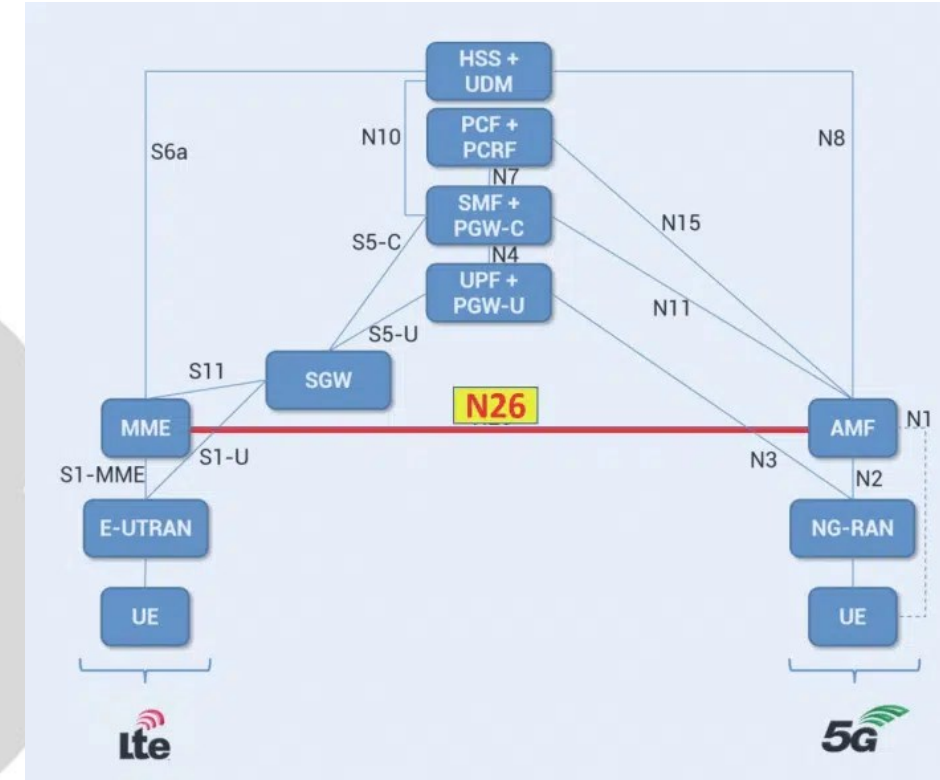
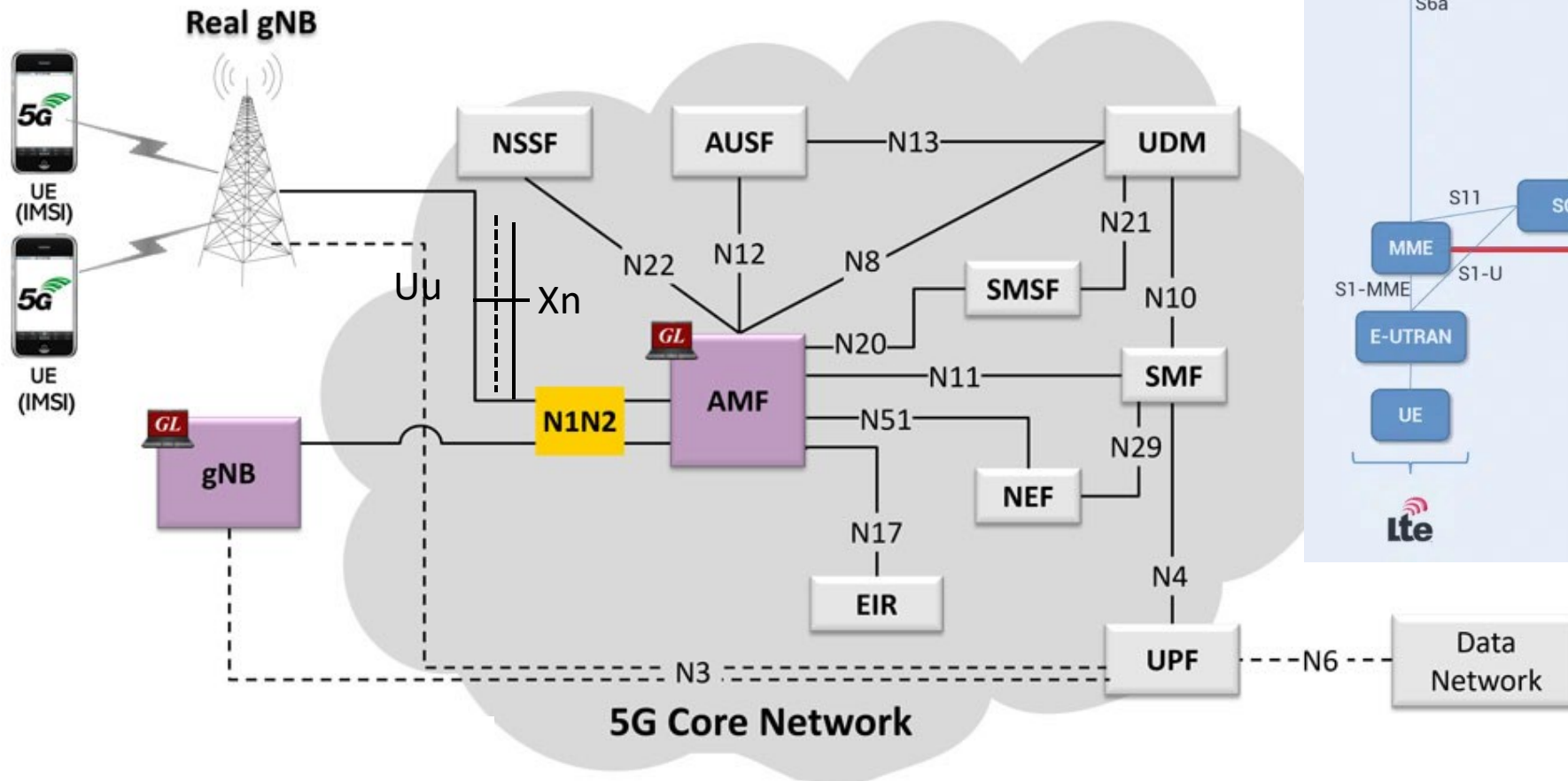
~ Leonardo da Vinci ~

5G E2E QoS Call Flow

60min

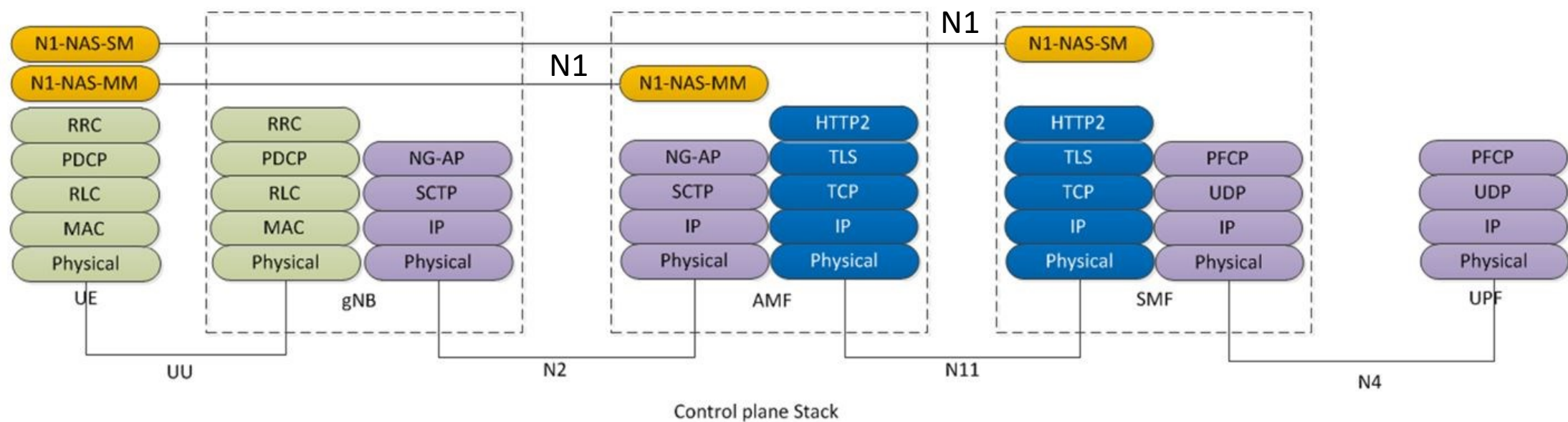


5G SA Network Architecture

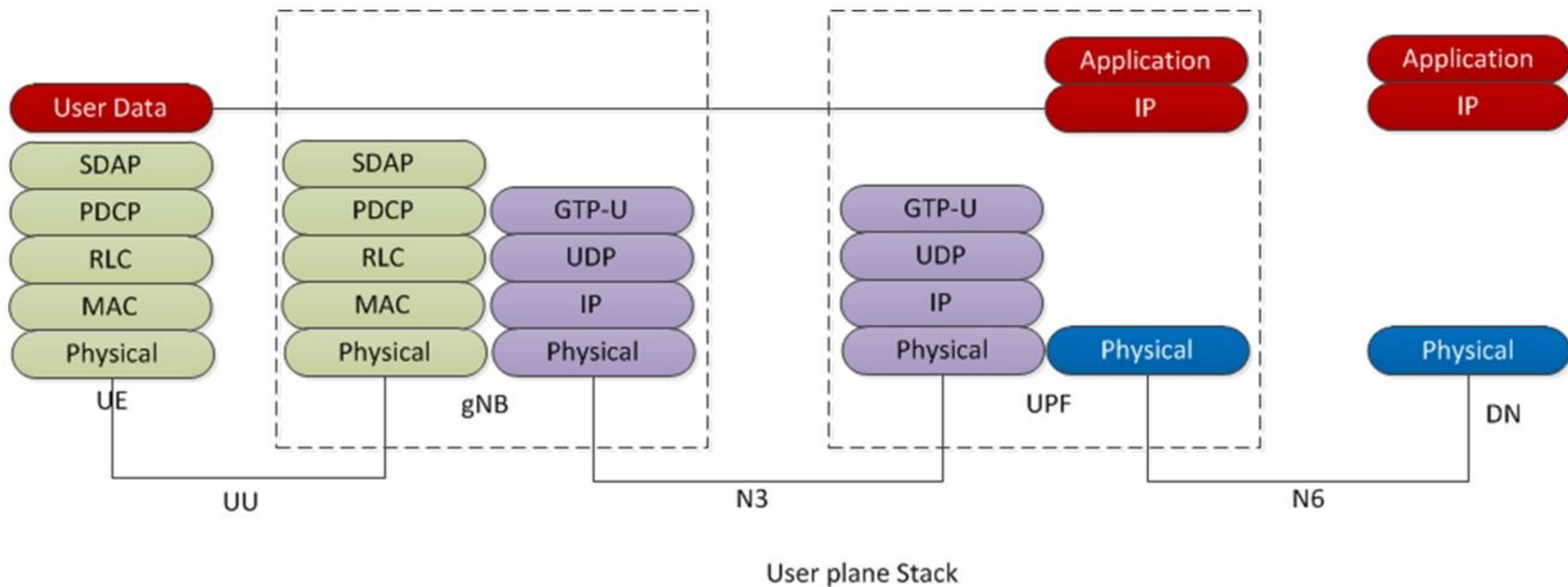


Source:
<https://www.5gworldpro.com/blog/2020/11/17/is-interface-n26-important-in-5g/>

5G E2E: Uu, N1, N2



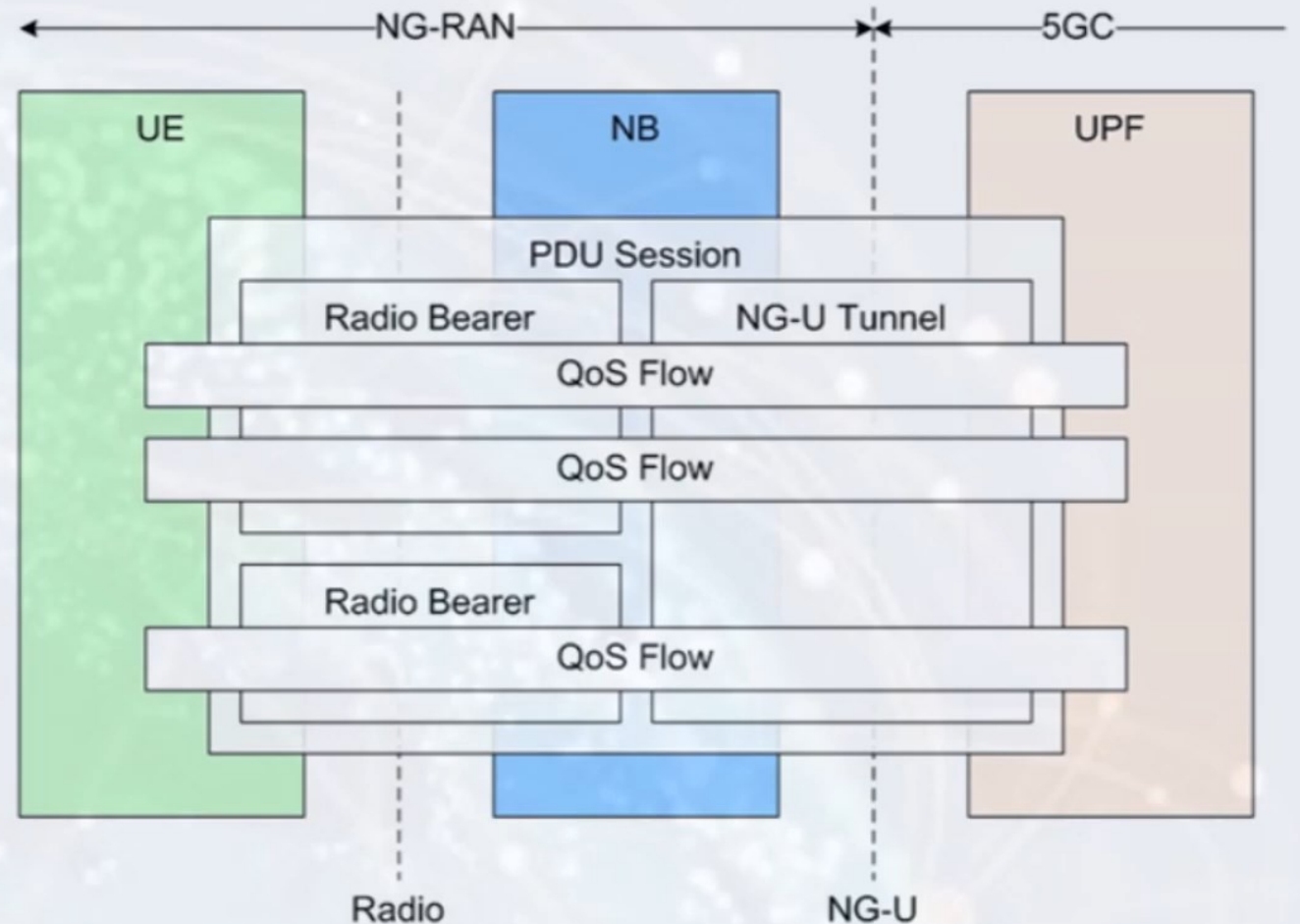
5G E2E: Uu, N3



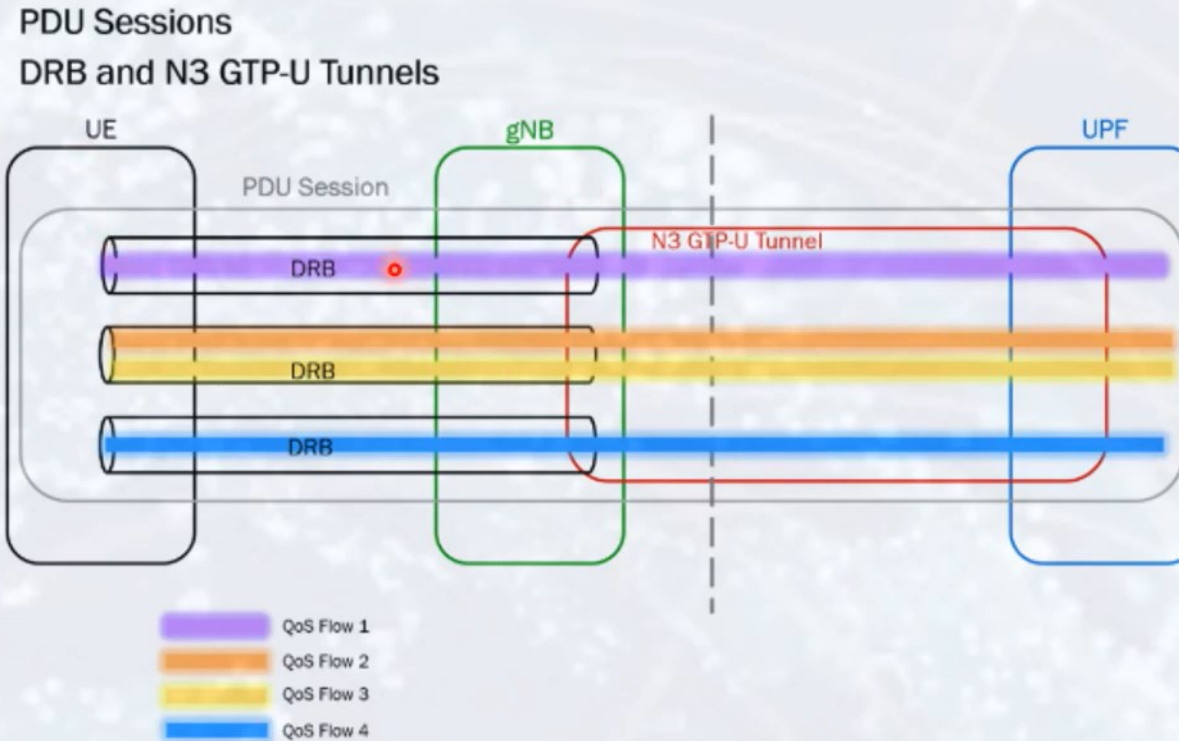
5G QoS Architecture

Types of QoS Flow

- QoS Flow with GBR
- QoS Flow with NGBR
- Delay Critical GBR
- Reflective QoS



5G QoS Architecture



A PDU session may contain multiple QoS flows and several DRBs but only a single N3 GTP-U tunnel. A DRB may transport one or more QoS flows.

Data Radio Bearer

~~DRB : Dedicated Radio Bearer~~

GTP : GPRS Tunnel Protocol

5G QoS Architecture

<https://clcnetwork.wordpress.com/2020/04/19/5g-core-pdu-session-and-qos-part-2/>

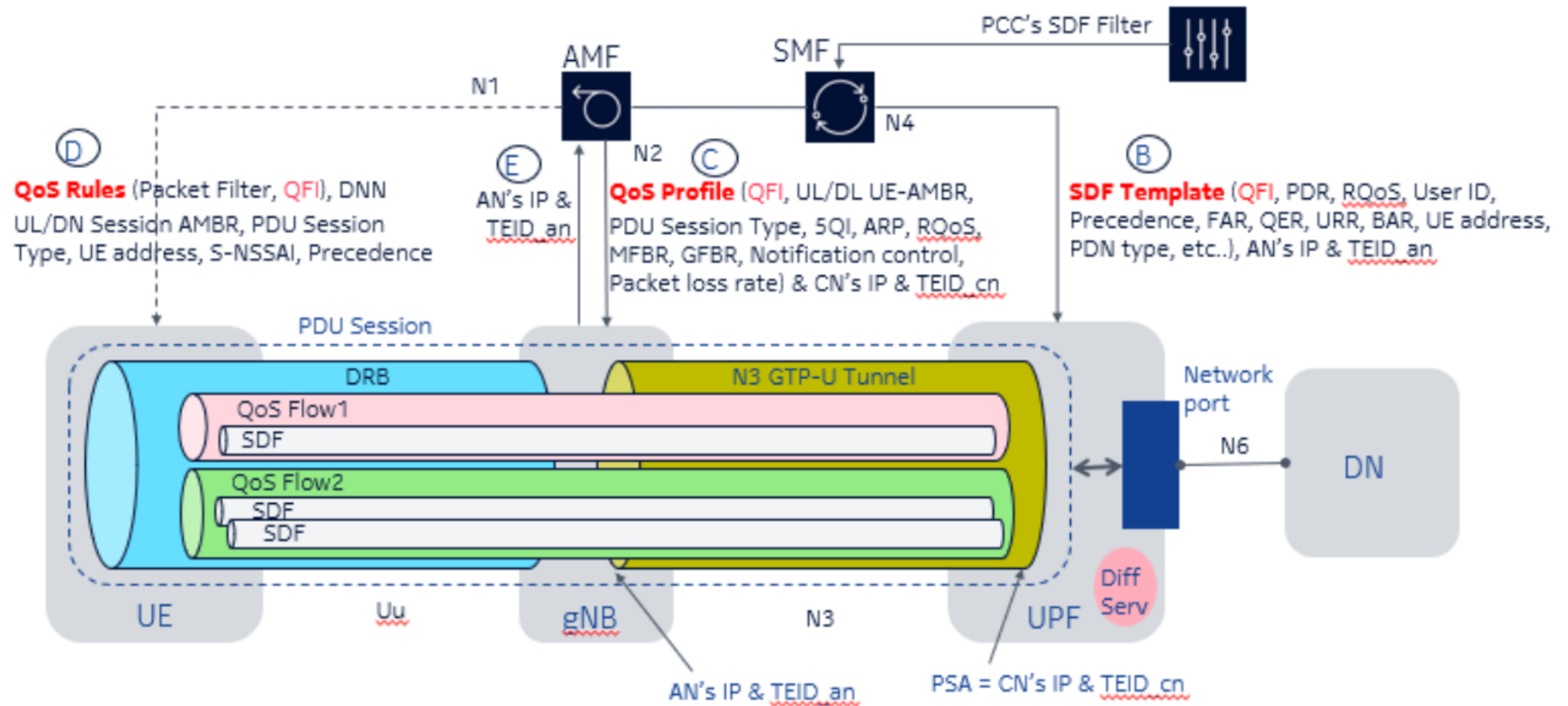


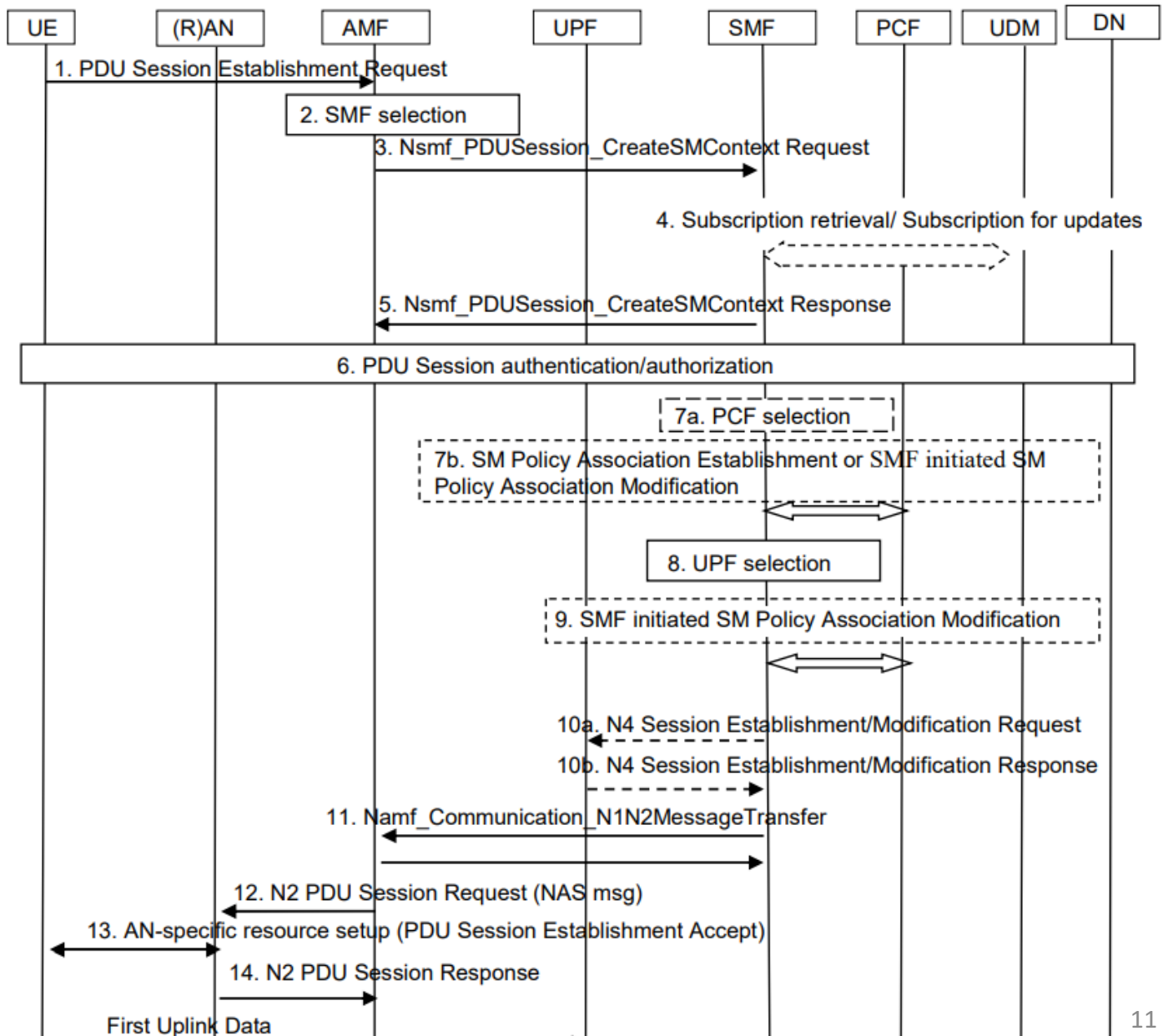
Figure 1 – SDF Template, QoS Profile and QoS Rule for UPF, gNB and UE respectively

5G QoS Call Procedures

<https://clcnetwork.wordpress.com/2020/04/19/5g-core-pdu-session-and-qos-part-2/>

- In order for a PDU session and a default QoS Flow to be established along the UE, gNB to UPF to carry SDFs (Service Data Flows) or traffic, a lot of IEs (Information Elements) are exchanged among AMF, SMF, UE, gNB and UPF as shown in the Figure 1:
- The SMF directly and indirectly (via the AMF and the gNB) sends the below QoS constructs for the PDU Session and the default QoS Flow Establishment:
 - SDF Template to UPF
 - QoS Profile to gNB
 - QoS Rule to UE
- Beside setting up the PDU Session and the default QoS Flow comprising a non-GBR Data Radio Bearer (DRB) between the UE and the gNB and the N3 GTP-U tunnel between gNB and UPF, the above IEs in the Figure 1 are also used for QoS processing and enforcement of the UE's traffic such as:
 - Classification — Classify or detect the UE's uplink (UL) and downlink (DL) traffic onto the appropriate QoS Flows within the UE's PDU session
 - Queuing and Scheduling — Enforce GBR and non-GBR UE traffic such as UL/DL bandwidth, latency, traffic priority etc...
 - Marking/Remarking — Mark the mobile traffic when it leaves the 5GC to the DN so that the priority and QoS of the traffic can be honored in the DN

5G QoS Call Procedures



5G QoS Call Procedures

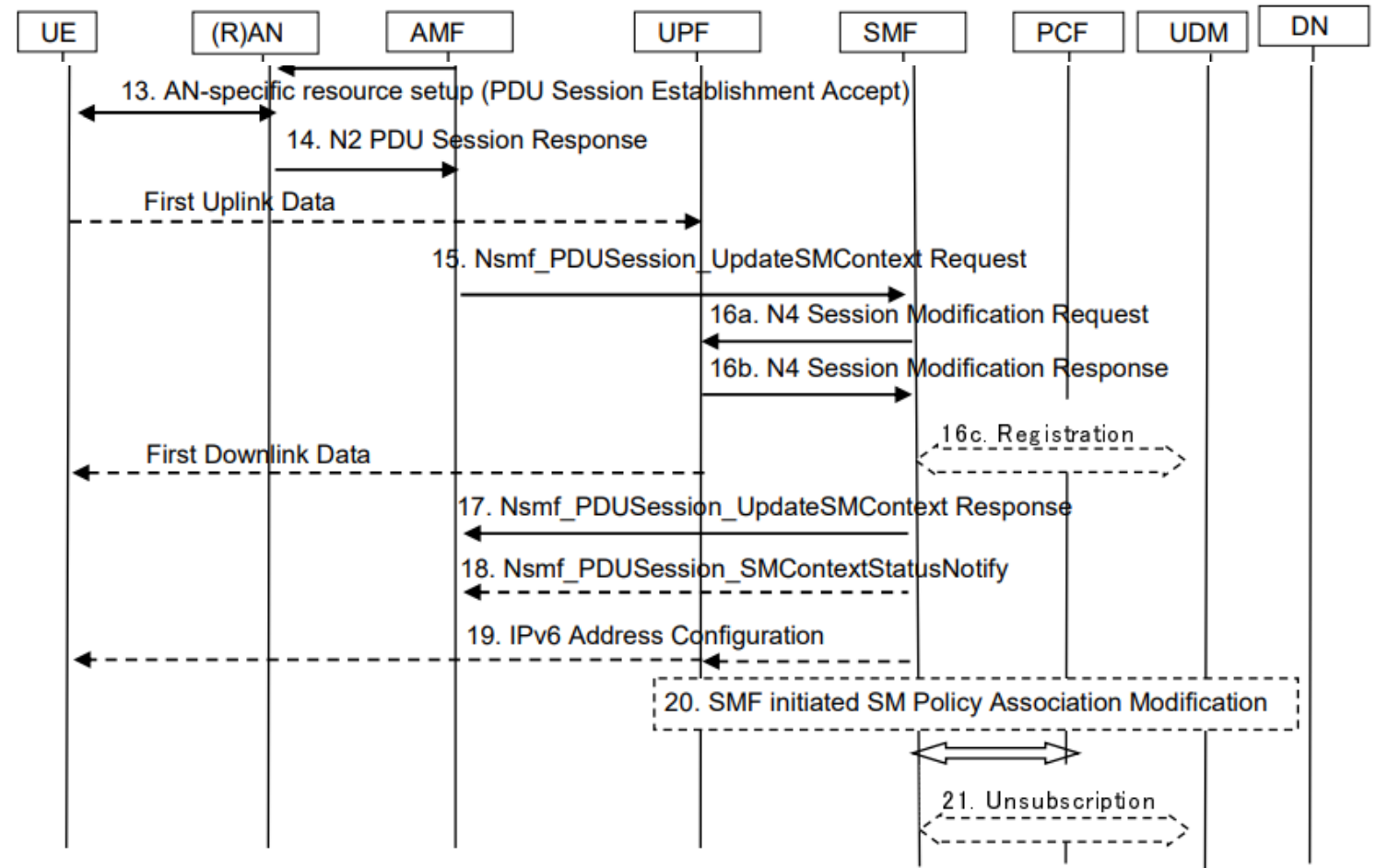


Figure 4.3.2.2.1-1: UE-requested PDU Session Establishment for non-roaming and roaming with local breakout

5G QoS Callflow in 5GCsim

- 5G SA
- 1 PDUSession
- With default 5QI=9, ping 10s
- Save E2E logs

2022_0411--5GCsimlog
Reg ok PDU ok Ping is less
but ok.pcap

5G QoS Callflow – after Registration

2022_0411--5Gsimlog Reg ok PDU ok Ping is less but ok.pcapng

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ngap||gtp 5G_ngap_sctp NGAP S1AP

Destination Address	Protocol	5QI	5G-TMSI	Info
192.168.50.100	NGAP			InitialContextSetupResponse
192.168.50.100	NGAP/NAS-5GS			UplinkNASTransport, Registration complete
192.168.50.100	NGAP/NAS-5GS			UplinkNASTransport, UL NAS transport, PDU session establishment request
192.168.50.102	NGAP/NAS-5GS	9		SACK (Ack=13, Arwnd=25165824) , PDUSessionResourceSetupRequest, DL NAS transport, PDU session establish
192.168.50.100	NGAP			PDUSessionResourceSetupResponse
172.16.0.1	192.168.50.101,22.22.22.22	GTP <ICMP>		Echo (ping) request id=0xc641, seq=0/0, ttl=10 (reply in 2005)
22.22.22.22	192.168.50.102,172.16.0.1	GTP <ICMP>		Echo (ping) reply id=0xc641, seq=0/0, ttl=64 (request in 2004)
172.16.0.1	192.168.50.101,22.22.22.22	GTP <ICMP>		Echo (ping) request id=0xc641, seq=2/512, ttl=10 (reply in 2007)
22.22.22.22	192.168.50.102,172.16.0.1	GTP <ICMP>		Echo (ping) reply id=0xc641, seq=2/512, ttl=64 (request in 2006)
172.16.0.1	192.168.50.101,22.22.22.22	GTP <ICMP>		Echo (ping) request id=0xc641, seq=3/768, ttl=10 (reply in 2009)
22.22.22.22	192.168.50.102,172.16.0.1	GTP <ICMP>		Echo (ping) reply id=0xc641, seq=3/768, ttl=64 (request in 2008)
172.16.0.1	192.168.50.101,22.22.22.22	GTP <ICMP>		Echo (ping) request id=0xc641, seq=4/1024, ttl=10 (reply in 2011)
22.22.22.22	192.168.50.102,172.16.0.1	GTP <ICMP>		Echo (ping) reply id=0xc641, seq=4/1024, ttl=64 (request in 2010)
172.16.0.1	192.168.50.101,22.22.22.22	GTP <ICMP>		Echo (ping) request id=0xc641, seq=5/1280, ttl=10 (reply in 2015)
22.22.22.22	192.168.50.102,172.16.0.1	GTP <ICMP>		Echo (ping) reply id=0xc641, seq=5/1280, ttl=64 (request in 2014)
172.16.0.1	192.168.50.101,22.22.22.22	GTP <ICMP>		Echo (ping) request id=0xc641, seq=6/1536, ttl=10 (reply in 2019)
22.22.22.22	192.168.50.102,172.16.0.1	GTP <ICMP>		Echo (ping) reply id=0xc641, seq=6/1536, ttl=64 (request in 2018)
172.16.0.1	192.168.50.101,22.22.22.22	GTP <ICMP>		Echo (ping) request id=0xc641, seq=7/1792, ttl=10 (reply in 2021)
22.22.22.22	192.168.50.102,172.16.0.1	GTP <ICMP>		Echo (ping) reply id=0xc641, seq=7/1792, ttl=64 (request in 2020)
172.16.0.1	192.168.50.101,22.22.22.22	GTP <ICMP>		Echo (ping) request id=0xc641, seq=8/2048, ttl=10 (reply in 2023)
22.22.22.22	192.168.50.102,172.16.0.1	GTP <ICMP>		Echo (ping) reply id=0xc641, seq=8/2048, ttl=64 (request in 2022)
172.16.0.1	192.168.50.101,22.22.22.22	GTP <ICMP>		Echo (ping) request id=0xc641, seq=9/2304, ttl=10 (reply in 2026)
22.22.22.22	192.168.50.102,172.16.0.1	GTP <ICMP>		Echo (ping) reply id=0xc641, seq=9/2304, ttl=64 (request in 2025)
192.168.50.100	NGAP/NAS-5GS	0 (0x00000000)		UplinkNASTransport, Deregistration request (UE originating)
192.168.50.102	NGAP/NAS-5GS			SACK (Ack=15, Arwnd=25165824) , DownlinkNASTransport, Deregistration accept (UE originating)
192.168.50.102	NGAP			UEContextReleaseCommand
192.168.50.100	NGAP			UEContextReleaseComplete

< >

> Frame 10: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface unknown, id 0
> Ethernet II, Src: Meiko_02:06:94 (00:00:fc:02:06:94), Dst: VMware_89:8a:28 (00:0c:29:89:8a:28)
> Internet Protocol Version 4, Src: 192.168.50.102, Dst: 192.168.50.100
> Stream Control Transmission Protocol, Src Port: 38412 (38412), Dst Port: 38412 (38412)
> NG Application Protocol (NGSetupRequest)

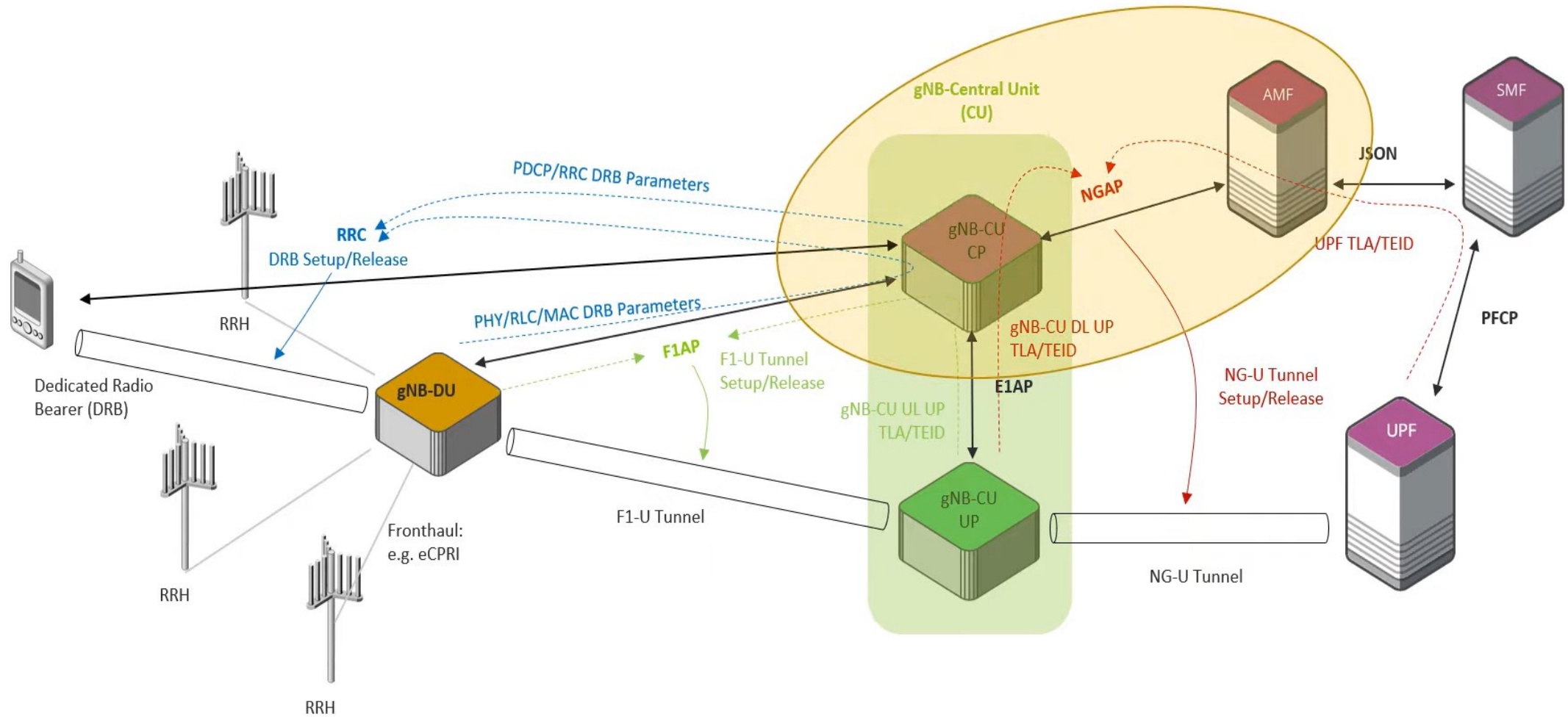
0000 00 0c 29 89 8a 2 ^
0010 00 60 00 01 40 0
0020 32 64 96 0c 96 0
0030 00 3d 85 c1 98 a v
< >
Frame (110 bytes) Bitstrin

GPWS Tunneling Protocol: Protocol Packets: 2082 · Displayed: 47 (2.3%) Profile: Default

在 這裡 輸入 文字 來 搜尋

上午 10:27 2024/5/19

Protocols Involved in Establishing UP Transport



5G QoS Callflow – PDUSessionEst /NAS_MM

5Gcsimlog Reg ok PDU ok Ping is less but ok.pcapng

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Destination Address	Protocol	5QI	5G-TMSI	Info
192.168.50.100	NGAP			InitialContextSetupResponse
192.168.50.100	NGAP/NAS-5GS			UplinkNASTransport, Registration complete
192.168.50.100	NGAP/NAS-5GS			UplinkNASTransport, UL NAS transport, PDU session establishment request
192.168.50.102	NGAP/NAS-5GS	9		SACK (Ack=13, Arwnd=25165824) , PDUSessionResourceSetupRequest, DL NAS transport, PDU session establish
192.168.50.100	NGAP			PDUSessionResourceSetupResponse

id: id-NAS-PDU (38)
criticality: reject (0)
value
NAS-PDU: 7e02570ace82027e006701000a2e0101c1ffff91280100120181250908696e7465726e6574
Non-Access-Stratum 5GS (NAS)PDU
Security protected NAS 5GS message
Plain NAS 5GS Message
Extended protocol discriminator: 5G mobility management messages (126)
0000 = Spare Half Octet: 0
.... 0000 = Security header type: Plain NAS message, not security protected (0)
Message type: UL NAS transport (0x67)
0000 = Spare Half Octet: 0
Payload container type
.... 0001 = Payload container type: N1 SM information (1)
Payload container
Length: 10
Plain NAS 5GS Message
PDU session identity 2 - PDU session ID
Element ID: 0x12
PDU session identity: PDU session identity value 1 (1)
Request type
1000 = Element ID: 0x8-
.... .001 = Request type: Initial request (1)
DNN
Element ID: 0x25
Length: 9
DNN: internet
Item 3: id-UserLocationInformation

NAS_SM

0000 00 0c 29 89 8a 28
0010 00 80 03 2b 40 00
0020 32 64 96 0c 96 0c
0030 00 60 85 c1 98 ae
0040 40 4c 00 00 04 00
0050 03 00 26 00 26 25
0060 01 00 0a 2e 01 01
0070 25 09 08 69 6e 74
0080 00 f1 10 01 00 00

Frame (142 bytes) Bitstring

GPRS Tunneling Protocol: Protocol Packets: 2082 · Displayed: 47 (2.3%) Profile: Default

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上午 10:41 2024/5/19

5G QoS Callflow – PDUSessionEst /NAS_SM

192.168.50.100	NGAP		initialContextSetupResponse
192.168.50.100	NGAP/NAS-5GS		UplinkNASTransport, Registration complete
192.168.50.100	NGAP/NAS-5GS		UplinkNASTransport, UL NAS transport, PDU session establishment request
192.168.50.102	NGAP/NAS-5GS	9	SACK (Ack=13, Arwnd=25165824) , PDUSessionResourceSetupRequest, DL NAS transport, PDU
192.168.50.100	NGAP		PDUSessionResourceSetupResponse

0000 = Spare Half Octet: 0

- ✓ Payload container type
 - 0001 = Payload container type: N1 SM information (1)
- ✓ Payload container
 - Length: 10
 - ✓ Plain NAS 5GS Message
 - Extended protocol discriminator: 5G session management messages (46)
 - PDU session identity: PDU session identity value 1 (1)
 - Procedure transaction identity: 1
 - Message type: PDU session establishment request (0xc1)
 - ✓ Integrity protection maximum data rate
 - Integrity protection maximum data rate for uplink: Full data rate (255)
 - Integrity protection maximum data rate for downlink: Full data rate (255)
 - ✓ PDU session type
 - 1001 = Element ID: 0x9-
 -001 = PDU session type: IPv4 (1)
 - ✓ 5GSM capability
 - Element ID: 0x28
 - Length: 1
 - 0... = Transfer of port management information containers (TPMIC): Not supported
 - .000 0... = Supported ATSSS steering functionalities and steering modes (ATSSS-ST): ATSSS not supported (0)
 -0.. = Ethernet PDN type in S1 mode (EPT-S1): Not supported
 -0. = Multi-homed IPv6 PDU session (MH6-PDU): Not supported
 -0 = Reflective QoS (RqoS): Not supported

> PDU session identity 2 - PDU session ID

> Request type

> DNN

> Item 3: id-UserLocationInformation

5G QoS Callflow – translations for human--1

UE to AMF:

- It is initial request to AMF
- To which DNN
- With which PDUSession ID
- PDUSessionEstRequest
- Max Data rate
- IPv4 or IPv6
- 5G SM Cap

5G QoS Callflow – PDUSession Resource --1

2022_0411--5Gsimlog Reg ok PDU ok Ping is less but ok.pcapng

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ngap|gtp

Destination Address	Protocol	5QI	5G-TMSI	Info
192.168.50.100	NGAP/NAS-5GS			UplinkNASTransport, UL NAS transport, PDU session establishment request
192.168.50.102	NGAP/NAS-5GS	9		SACK (Ack=13, Arwnd=25165824) , PDUSessionResourceSetupRequest, DL NAS transport, PDU session establish
192.168.50.100	NGAP			PDUSessionResourceSetupResponse

< | >

NG Application Protocol (PDUSessionResourceSetupRequest)

- NGAP-PDU: initiatingMessage (0)
 - initiatingMessage
 - procedureCode: id-PDUSessionResourceSetup (29)
 - criticality: reject (0)
 - value
 - PDUSessionResourceSetupRequest
 - protocolIEs: 4 items
 - Item 0: id-AMF-UE-NGAP-ID
 - Item 1: id-RAN-UE-NGAP-ID
 - Item 2: id-PDUSessionResourceSetupListSUReq
 - ProtocolIE-Field
 - id: id-PDUSessionResourceSetupListSUReq (74)
 - criticality: reject (0)
 - value
 - PDUSessionResourceSetupListSUReq: 1 item
 - Item 0
 - PDUSessionResourceSetupItemSUReq
 - pDUSessionID: 1
 - pDUSessionNAS-PDU [truncated]: 7e024a170591027e006801006b2e0101c21100090900063130010109090603f42403f424290501ac10000122040103...
 - s-NSSAI
 - pDUSessionResourceSetupRequestTransfer: 0000050082000a0c3b9aca00303b9aca00008b000a01f0c0a83265000000040086000100008a000240200...
 - Item 3: id-UEAggregateMaximumBitRate
 - ProtocolIE-Field
 - id: id-UEAggregateMaximumBitRate (110)
 - criticality: ignore (1)
 - value
 - UEAggregateMaximumBitRate
 - uAggregateMaximumBitRateDL: 1000000000bits/s
 - uAggregateMaximumBitRateUL: 1000000000bits/s

0000 00 00 fc 02 06 94
0010 01 20 00 00 40 00
0020 32 66 96 0c 96 0c
0030 00 10 85 c1 98 ae
0040 00 f0 f8 2a 0a a5
0050 00 80 db 00 00 04
0060 00 03 00 4a 00 80
0070 91 02 7e 00 68 01
0080 00 06 31 30 01 01
0090 05 01 ac 10 00 01
00a0 00 0c 52 01 01 09
00b0 0c 51 01 09 03 fe
00c0 51 01 09 06 fe fe
00d0 20 42 01 01 09 07
00e0 6e 65 74 12 01 40
00f0 00 0a 0c 3b 9a ca
0100 01 f0 c0 a8 32 65
0110 8a 00 02 40 20 00
0120 00 6e 40 0a 0c 3b

Frame (302 bytes) Unaligned

PDU Session Resource Setup Request (ngap.PDUSessionResourceSetupRequest_element), 219 byte(s) | Packets: 2082 · Displayed: 47 (2.3%) | Profile: Default

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上午 11:15
2024/5/19

5G QoS Callflow – PDUSession Resource --2

2022_0411--5Gcsimlog Reg ok PDU ok Ping is less but okpcapng

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ngap|gtp

| on Address | Protocol | 5QI | 5G-TMSI | Info |
|----------------|--------------|-----|---------|--|
| 192.168.50.100 | NGAP/NAS-5GS | | | UplinkNASTransport, UL NAS transport, PDU session establishment request |
| 192.168.50.102 | NGAP/NAS-5GS | 9 | | SACK (Ack=13, Arwnd=25165824) , PDUSessionResourceSetupRequest, DL NAS transport, PDU session establishment accept |
| 192.168.50.100 | NGAP | | | PDUSessionResourceSetupResponse |

PDUSessionResourceSetupItemSUReq

- pDUSessionID: 1
- pDUSessionNAS-PDU [truncated]: 7e024a170591027e006801006b2e0101c21100090900063130010109090603f42403f424290501ac1000012204010300
- s-NSSAI
 - sST: 01
 - sD: 030609
- pDUSessionResourceSetupRequestTransfer: 0000050082000a0c3b9aca00303b9aca00008b000a01f0c0a83265000000040086000100008a0002402000
- PDUSessionResourceSetupRequestTransfer
 - protocolIEs: 5 items
 - Item 0: id-PDUSessionAggregateMaximumBitRate
 - ProtocolIE-Field
 - id: id-PDUSessionAggregateMaximumBitRate (130)
 - criticality: reject (0)
 - value
 - PDUSessionAggregateMaximumBitRate
 - pDUSessionAggregateMaximumBitRateDL: 1000000000bits/s
 - pDUSessionAggregateMaximumBitRateUL: 1000000000bits/s
 - Item 1: id-UL-NGU-UP-TNLInformation
 - ProtocolIE-Field
 - id: id-UL-NGU-UP-TNLInformation (139)
 - criticality: reject (0)
 - value
 - UPTransportLayerInformation: gTPTunnel (0)
 - gTPTunnel
 - transportLayerAddress: c0a83265 [bit length 32, 1100 0000 1010 1000 0011 0010 0110 0101 decimal value 32]
TransportLayerAddress (IPv4): 192.168.50.101
gTP-TEID: 00000004
 - Item 2: id-PDUSessionType
 - Item 3: id-SecurityIndication

0000 00 00 fc 02 06 94
0010 01 20 00 00 40 00
0020 32 66 96 0c 96 0c
0030 00 10 85 c1 98 ae
0040 00 f0 f8 2a 0a a5
0050 00 80 db 00 00 04
0060 00 03 00 4a 00 80
0070 91 02 7e 00 68 01
0080 00 06 31 30 01 01
0090 05 01 ac 10 00 01
00a0 00 0c 52 01 01 09
00b0 0c 51 01 09 03 fe
00c0 51 01 09 06 fe fe
00d0 20 42 01 01 09 07
00e0 6e 65 74 12 01 40
00f0 00 0a 0c 3b 9a ca
0100 01 f0 c0 a8 32 65
0110 8a 00 02 40 20 00
0120 00 6e 40 0a 0c 3b

s-NSSAI (ngap.s-NSSAI_element), 5 byte(s)

Packets: 2082 · Displayed: 47 (2.3%)

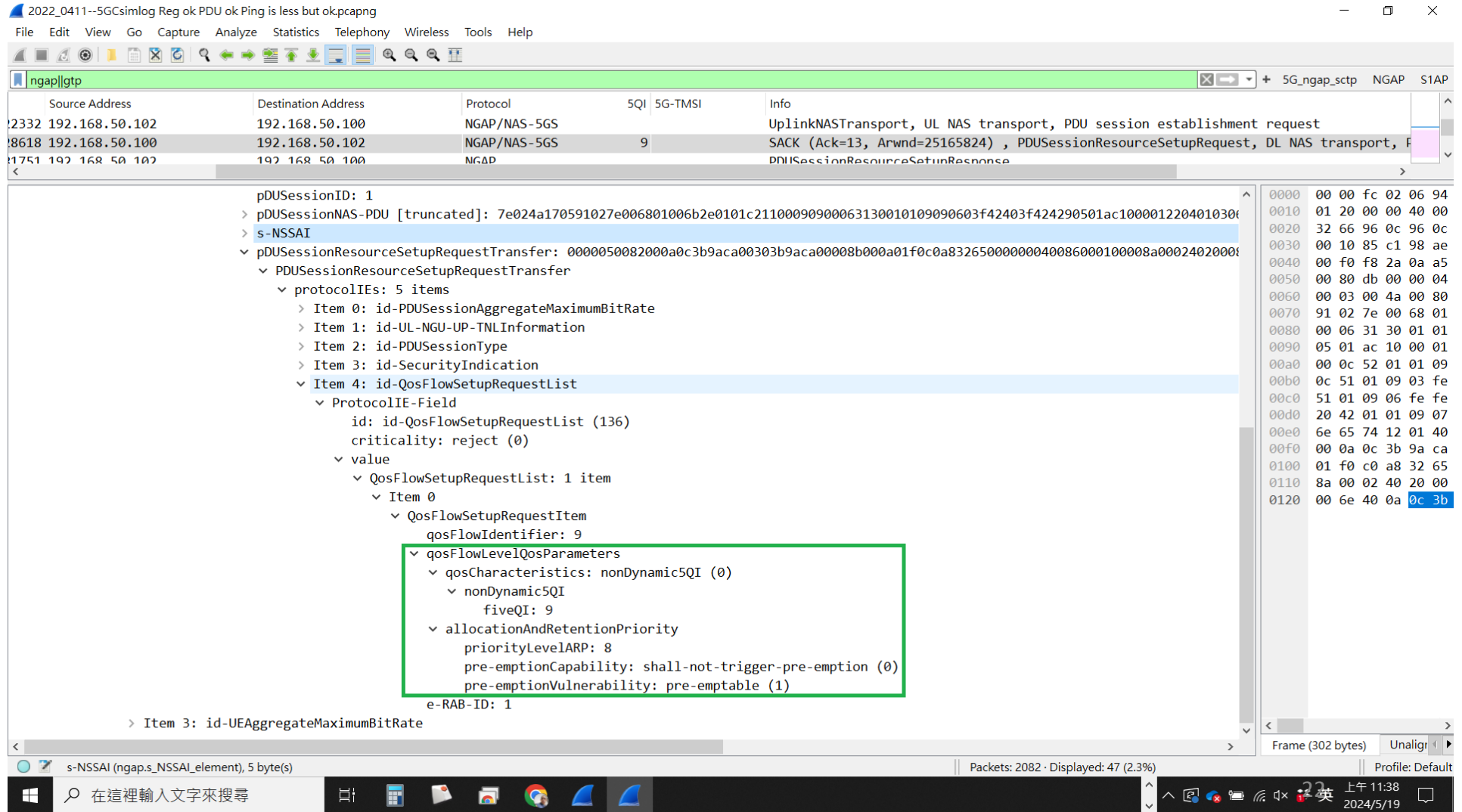
Frame (302 bytes) Unalign

Profile: Default

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上午 11:22
2024/5/19

5G QoS Callflow – PDUSession Resource --4



5G QoS Callflow – translations for human--2

AMF to gNB:

- UE AMBR
- PDUSession AMBR
- S-NSSAI of UE (which UE sent in Registration)
- UP gtp TEID, TLA
- QoS Flow Setup Request: QFI, e-RAB-ID, 5QI=9

5G QoS flows – Dynamic 5QI

5G NR Standardized QoS Identifier (5GQI) to QoS Characteristics Mapping

Packet Delay Budget (PDB)

BY AUTHOR · PUBLISHED JULY 25, 2019 · UPDATED JULY 25, 2019

Standardized 5G QoS Identifier (5QI) values are specified for services that are assumed to be frequently used in 5G networks and thus benefit from optimized signalling by using standardized QoS characteristics. Dynamically assigned 5QI values which require a signalling of QoS characteristics as part of the QoS profile can be used for services for which standardized 5QI values are not defined. The one-to-one mapping of standardized 5QI values to 5G QoS characteristics is specified in following table

| 5QI Value | Resource Type | Default Priority Level | Packet Delay Budget | Packet Error Rate | Default Maximum Data Burst Volume | Default Averaging Window | Example Services |
|-----------|---------------|------------------------|---------------------|-------------------|-----------------------------------|--------------------------|------------------|
| | | | | | (NOTE 2) | | |

5G QoS flows – Pre-emption

Allocation and Retention Priority (ARP)

The QoS parameter ARP contains information about the priority level, the pre-emption capability and the pre-emption vulnerability. The ARP priority level defines the relative importance of a resource request to allow in deciding whether a new QoS Flow may be accepted or needs to be rejected in the case of resource limitations (typically used for admission control of GBR traffic). It may also be used to decide which existing QoS Flow to pre-empt during resource limitations. ARP has following characteristics:

- The range of the ARP priority level is 1 to 15 with 1 as the highest level of priority
- The ARP priority levels 1-8 should only be assigned to resources for services that are authorized to receive prioritized treatment within an operator domain (i.e. that are authorized by the serving network)
- The ARP priority levels 9-15 may be assigned to resources that are authorized by the home network and thus applicable when a UE is roaming
- The ARP pre-emption capability defines whether a service data flow may get resources that were already assigned to another service data flow with a lower ARP priority level
- The ARP pre-emption capability and the ARP pre-emption vulnerability shall be either set to 'enabled' or 'disabled'
- The ARP pre-emption vulnerability defines whether a service data flow may lose the resources assigned to it in order to admit a service data flow with higher ARP priority level
- The ARP pre-emption vulnerability of the QoS Flow which the default QoS rule is associated with should be set appropriately to minimize the risk of unnecessary release of this QoS Flow

5G QoS Callflow – PDUSessionEstAccept--1

2022_0411--5Gsimlog Reg ok PDU ok Ping is less but ok.pcapng

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ngap|gtp 5G_ngap_sctp NGAP S1AP

| Address | Protocol | 5QI | 5G-TMSI | Info |
|---------|--------------|-----|---------|---|
| 0.100 | NGAP/NAS-5GS | | | UplinkNASTransport, UL NAS transport, PDU session establishment request |
| 0.102 | NGAP/NAS-5GS | 9 | | SACK (Ack=13, Arwnd=25165824) , PDUSessionResourceSetupRequest, DL NAS transport, PDUSessionResourceSetupItemSUReq |
| 0.100 | NGAP | | | PDUSessionResourceSetupResponse |

PDUSessionResourceSetupItemSUReq

- pDUSessionID: 1
- PDUSessionNAS-PDU [truncated]: 7e024a170591027e006801006b2e0101c21100090900063130010109090603f42403f424290501ac1000012204010300
- Non-Access-Stratum 5GS (NAS)PDU
 - Security protected NAS 5GS message
 - Plain NAS 5GS Message
 - Extended protocol discriminator: **5G mobility management messages (126)**
 - 0000 = Spare Half Octet: 0
 - 0000 = Security header type: Plain NAS message, not security protected (0)
 - Message type: DL NAS transport (0x68)
 - 0000 = Spare Half Octet: 0
 - Payload container type
 - 0001 = Payload container type: **N1 SM information (1)**
 - Payload container
 - Length: 107
 - Plain NAS 5GS Message
 - Extended protocol discriminator: **5G session management messages (46)**
 - PDU session identity: PDU session identity value 1 (1)
 - Procedure transaction identity: 1
 - Message type: PDU session establishment accept (0xc2)**
 - .001 = Selected SSC mode: SSC mode 1 (1)
 - PDU session type - Selected PDU session type
 - QoS rules - Authorized QoS rules
 - Session-AMBR
 - PDU address
 - S-NSSAI
 - Mapped EPS bearer contexts
 - QoS flow descriptions - Authorized
 - DNN

0000 00 00 fc 02 06 94
0010 01 20 00 00 40 00
0020 32 66 96 0c 96 0c
0030 00 10 85 c1 98 ae
0040 00 f0 f8 2a 0a a5
0050 00 80 db 00 00 04
0060 00 03 00 4a 00 80
0070 91 02 7e 00 68 01
0080 00 06 31 30 01 01
0090 05 01 ac 10 00 01
00a0 00 0c 52 01 01 09
00b0 0c 51 01 09 03 fe
00c0 51 01 09 06 fe fe
00d0 20 42 01 01 09 07
00e0 6e 65 74 12 01 40
00f0 00 0a 0c 3b 9a ca
0100 01 f0 c0 a8 32 65
0110 8a 00 02 40 20 00
0120 00 6e 40 0a 0c 3b

allocationAndRetentionPriority (ngap.allocationAndRetentionPriority_element), 1 byte(s) | Packets: 2082 · Displayed: 47 (2.3%) | Profile: Default

5G QoS Callflow – PDUSessionEstAccept--2

2022_0411--5Gcsimlog Reg ok PDU ok Ping is less but ok.pcapng

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ngap||gtp

| Address | Protocol | 5QI | 5G-TMSI | Info |
|---------|--------------|-----|---------|--|
| 0.100 | NGAP/NAS-5GS | | | UplinkNASTransport, UL NAS transport, PDU session establishment request |
| 0.102 | NGAP/NAS-5GS | 9 | | SACK (Ack=13, Arwnd=25165824) , PDUSessionResourceSetupRequest, DL NAS transport, PDU session establishment accept |
| 0.100 | NGAP | | | PDUSessionResourceSetupResponse |

Procedure transaction identity: 1
Message type: PDU session establishment accept (0xc2)
.001 = Selected SSC mode: SSC mode 1 (1)
PDU session type - Selected PDU session type
....001 = PDU session type: IPv4 (1)
QoS rules - Authorized QoS rules
Length: 9
QoS rule 1
QoS rule identifier: 9
Length: 6
001. = Rule operation code: Create new QoS rule (1)
...1 = DQR: The QoS rule is the default QoS rule
.... 0001 = Number of packet filters: 1
Packet filter 1
..11 = Packet filter direction: Bidirectional (3)
.... 0000 = Packet filter identifier: 0
Length: 1
Packet filter component 1
Packet filter component type: Match-all type (1)
QoS rule precedence: 9
0... = Spare: 0
.0.. = Spare: 0
..00 1001 = Qos flow identifier: 9
Session-AMBR
Length: 6
Unit for Session-AMBR for downlink: value is incremented in multiples of 16 Kbps (3)
Session-AMBR for downlink: 1000000 Kbps (62500)
Unit for Session-AMBR for uplink: value is incremented in multiples of 16 Kbps (3)
Session-AMBR for uplink: 1000000 Kbps (62500)

0000 00 00 fc 02 06 94
0010 01 20 00 00 40 00
0020 32 66 96 0c 96 0c
0030 00 10 85 c1 98 ae
0040 00 f0 f8 2a 0a a5
0050 00 80 db 00 00 04
0060 00 03 00 4a 00 80
0070 91 02 7e 00 68 01
0080 00 06 31 30 01 01
0090 05 01 ac 10 00 01
00a0 00 0c 52 01 01 09
00b0 0c 51 01 09 03 fe
00c0 51 01 09 06 fe fe
00d0 20 42 01 01 09 07
00e0 6e 65 74 12 01 40
00f0 00 0a 0c 3b 9a ca
0100 01 f0 c0 a8 32 65
0110 8a 00 02 40 20 00
0120 00 6e 40 0a 0c 3b

allocationAndRetentionPriority (ngap.allocationAndRetentionPriority_element), 1 byte(s)

Packets: 2082 · Displayed: 47 (2.3%)

Profile: Default

5G QoS flows – Authorized QoS

4.2.6.6 Authorized QoS

4.2.6.6.1 General

3GPP TS 29.512 version 16.5.0 Release 16

The PCF shall provision the authorized QoS. The authorized QoS may apply to a PCC rule or to a PDU session.

- When the authorized QoS applies to a PCC rule, it shall be provisioned within the corresponding PCC rule as defined in subclause 4.2.6.6.2.
- When the authorized QoS for a PCC rule with a GBR QCI is candidate for resource sharing an instruction on the allowed sharing may be provisioned as defined in subclause 4.2.6.2.8.
- When the authorized QoS applies to a PDU session, it shall be provisioned as defined in subclause 4.2.6.3.1.
- When the authorized QoS applies to the default QoS flow, it shall be provisioned as defined in subclause 4.2.6.3.1.
- When the authorized QoS applies to an explicitly signalled QoS Characteristics, it shall be provisioned as defined in subclause 4.2.6.6.3.
- When the authorized QoS applies to the Reflective QoS, it shall be provisioned as defined in subclause 4.2.6.5.7.

The authorized QoS provides appropriate values for the resources to be enforced. The authorized QoS for a PCC rule is a request for allocating the corresponding resources. The Provisioning of authorized QoS per PCC rule is a part of PCC rule provisioning procedure.

If the SMF cannot allocate any of the resources as authorized by the PCF, the SMF informs the PCF and acts as described in subclauses 4.2.3.16 and 4.2.4.15.

5G QoS Callflow – PDUSessionEstAccept-- 3

2022_0411--5Gsimlog Reg ok PDU ok Ping is less but ok.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ngap||gtp 5G_ngap_sctp NGAP S1AP

| Address | Protocol | 5QI | 5G-TMSI | Info |
|---------|--------------|-----|---------|---|
| 0.100 | NGAP/NAS-5GS | | | UplinkNASTransport, UL NAS transport, PDU session establishment request |
| 0.102 | NGAP/NAS-5GS | 9 | | SACK (Ack=13, Arwnd=25165824) , PDUSessionResourceSetupRequest, DL NAS transport, PDU session establishment accept |
| 0.100 | NGAP | | | PDUSessionResourceSetupResponse |

Message type: PDU session establishment accept (0xc2)
.001 = Selected SSC mode: SSC mode 1 (1)
> PDU session type - Selected PDU session type
> QoS rules - Authorized QoS rules
> Session-AMBR
v PDU address
Element ID: 0x29
Length: 5
.... 0... = SMF's IPv6 link local address (SI6LLA): Absent
.... .001 = PDU session type: IPv4 (1)
PDU address information: 172.16.0.1
v S-NSSAI
Element ID: 0x22
Length: 4
Slice/service type (SST): eMBB (1)
Slice differentiator (SD): 198153
v Mapped EPS bearer contexts
Element ID: 0x75
Length: 45
> Mapped EPS bearer context 1
> Mapped EPS bearer context 2
> Mapped EPS bearer context 3
> QoS flow descriptions - Authorized
v DNN
Element ID: 0x25
Length: 9
DNN: internet
> PDU session identity 2 - PDU session ID
> s-NSSAI

小算盤

程式設計人員

198,153

HEX 3 0609
DEC 198,153
OCT 603 011
BIN 0011 0000 0110 0000 1001
QWORD MS Mv

位元 位元移位

A << >> CE ☒
B () % ÷
C 7 8 9 ×
D 4 5 6 -
E 1 2 3 +
F +/- 0 . =

0000 00 00 fc 02 06 94
0010 01 20 00 00 40 00
0020 32 66 96 0c 96 0c
0030 00 10 85 c1 98 ae
0040 00 f0 f8 2a 0a a5
0050 00 80 db 00 00 04
0060 00 03 00 4a 00 80
0070 91 02 7e 00 68 01
0080 00 06 31 30 01 01
0090 05 01 ac 10 00 01
00a0 00 0c 52 01 01 09
00b0 0c 51 01 09 03 fe
00c0 51 01 09 06 fe fe
00d0 20 42 01 01 09 07
00e0 6e 65 74 12 01 40
00f0 00 0a 0c 3b 9a ca
0100 01 f0 c0 a8 32 65
0110 8a 00 02 40 20 00
0120 00 6e 40 0a 0c 3b

Text item (text), 7 byte(s)

Packets: 2082 · Displayed: 47 (2.3%)

Profile: Default

在 這裡 輸入 文字 來 搜尋

下午 05:22
2024/5/19

5G QoS Callflow – PDU sessionEstAccept-- 4

2022_0411--5Gsimlog Reg ok PDU ok Ping is less but ok.pcapng

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ngap|gtp

| Address | Protocol | 5QI | 5G-TMSI | Info |
|---------|--------------|-----|---------|---|
| 0.100 | NGAP/NAS-5GS | | | UplinkNASTransport, UL NAS transport, PDU session establishment request |
| 0.102 | NGAP/NAS-5GS | 9 | | SACK (Ack=13, Arwnd=25165824) , PDUSessionResourceSetupRequest, DL NAS transport, PDU session establishment accept |
| 0.100 | NGAP | | | PDUSessionResourceSetupResponse |

< >

- > PDU session type - Selected PDU session type
- > QoS rules - Authorized QoS rules
- > Session-AMBR
- > PDU address
- > S-NSSAI
- > Mapped EPS bearer contexts
- ✓ **QoS flow descriptions - Authorized**
 - Element ID: 0x79
 - Length: 9
 - ✓ QoS flow description 1 - 5QI - EPS bearer identity
 - ..00 1001 = **Qos flow identifier: 9**
 - 001. = Operation code: **Create new QoS flow description (1)**
 - .1.. = E bit: 1
 - ..00 0010 = Number of parameters: 2
 - ✓ Parameter 1
 - Parameter identifier: 5QI (1)
 - Length: 1
 - 5QI: 9**
 - ✓ Parameter 2
 - Parameter identifier: EPS bearer identity (7)
 - Length: 1
 - 0001 = **EPS bearer identity: 1**
 - > DNN
 - ✓ PDU session identity 2 - PDU session ID
 - Element ID: 0x12
 - PDU session identity: **PDU session identity value 1 (1)**
 - > s-NSSAI
 - > pDUSessionResourceSetupRequestTransfer: 0000050082000a0c3b9aca00303b9aca00008b000a01f0c0a83265000000040086000100008a0002402000
- > Item 3: id-UEAggregateMaximumBitRate

0000 00 00 fc 02 06 94
0010 01 20 00 00 40 00
0020 32 66 96 0c 96 0c
0030 00 10 85 c1 98 ae
0040 00 f0 f8 2a 0a a5
0050 00 80 db 00 00 04
0060 00 03 00 4a 00 80
0070 91 02 7e 00 68 01
0080 00 06 31 30 01 01
0090 05 01 ac 10 00 01
00a0 00 0c 52 01 01 09
00b0 0c 51 01 09 03 fe
00c0 51 01 09 06 fe fe
00d0 20 42 01 01 09 07
00e0 6e 65 74 12 01 40
00f0 00 0a 0c 3b 9a ca
0100 01 f0 c0 a8 32 65
0110 8a 00 02 40 20 00
0120 00 6e 40 0a 0c 3b

Text item (text), 7 byte(s)

Packets: 2082 · Displayed: 47 (2.3%)

Frame (302 bytes) Unalign

Profile: Default

在 這裡 輸入 文字 來 搜尋

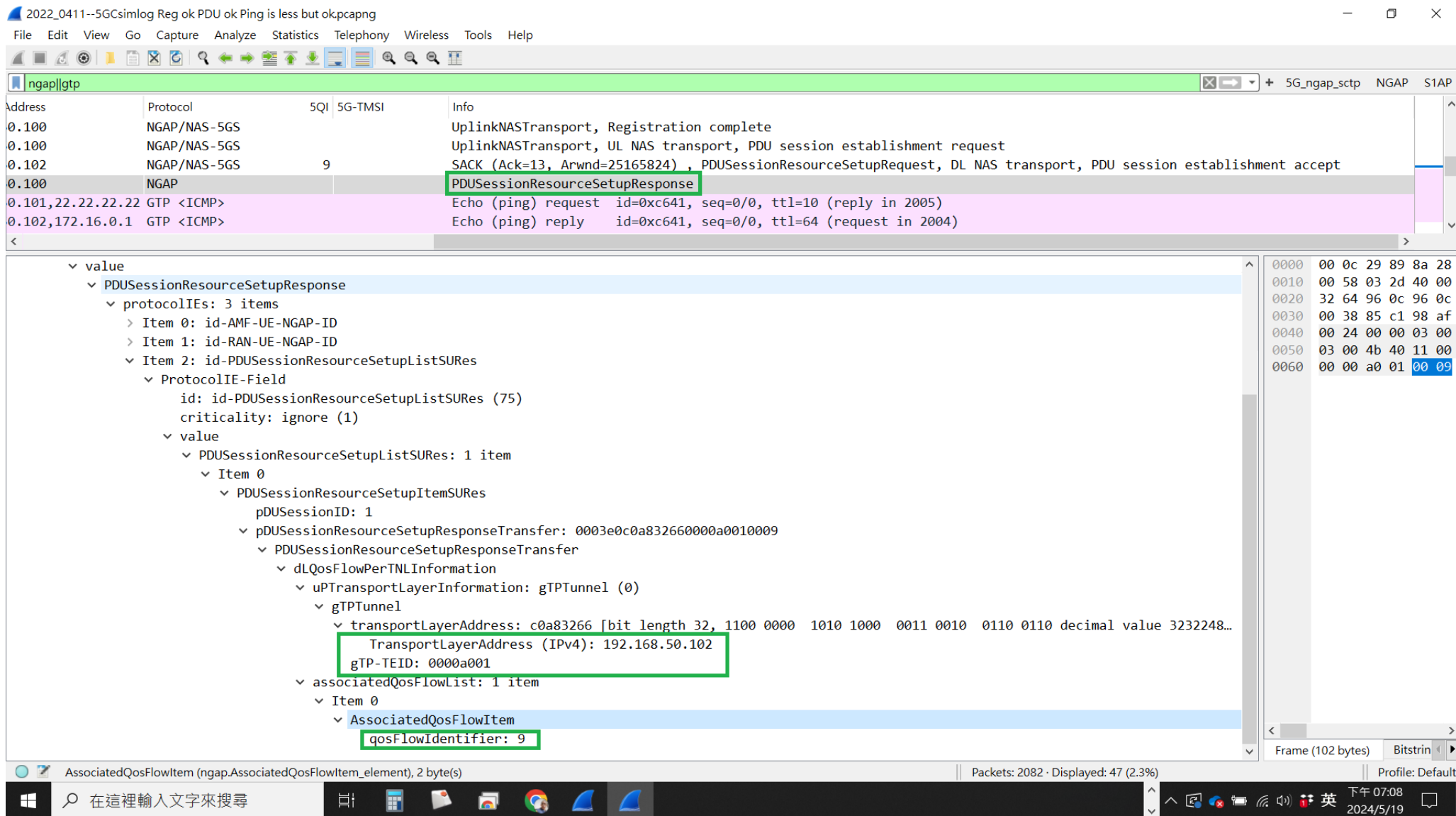
下午 05:40
2024/5/19

5G QoS Callflow – translations for human--3

AMF to UE:

- Accept the PDUSessionEstRequest
- Send Authorized QoS rules to UE: to create new QoS rule, packet filter, QoS Flow ID
- Session AMBR
- PDU IP address
- S-NSSAI, DNN
- Send QoS Flow description to UE: QoS Flow ID, 5QI, EPS bearer ID
- PDUSession ID

5G QoS Callflow – PDUSession ResourceRSP



5G QoS Callflow – translations for human--4

gNB to AMF:

- UP gtp TEID, TLA
- QoS Flow ID

5G QoS Callflow – UElog

2022_0411--UElog Reg ok
PDU ok Ping Complete.Isu

2022_0411--UElog Reg ok PDU ok Ping complete.Isu

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(((nas_5gs || nr-rrc) && !su.userid==1) || icmp) && !(nr5g.pkt_type == 13)

| No. | Time | User Label | User context id | 5QI | Info | type | UE Contention Resolution Identity | PreambleIndex | RAPID | UL |
|------|-------------------------------|------------|-----------------|-----|---|------|-----------------------------------|---------------|-------|----|
| 2143 | 2022-04-11 10:24:22.567838000 | uplink | 1 | | RRC Setup Request | | | | | |
| 2164 | 2022-04-11 10:24:22.613802000 | downlink | 1 | | RRC Setup | | | | | |
| 2170 | 2022-04-11 10:24:22.631859000 | uplink | 1 | | RRC Setup Complete (Registration request) | | | | | |
| 2190 | 2022-04-11 10:24:22.660801000 | downlink | 1 | | DL Information Transfer (Authentication request) | | | | | |
| 2194 | 2022-04-11 10:24:22.663850000 | uplink | 1 | | UL Information Transfer (Authentication response) | | | | | |
| 2237 | 2022-04-11 10:24:22.789793000 | downlink | 1 | | DL Information Transfer (Security mode command) | | | | | |
| 2238 | 2022-04-11 10:24:22.792855000 | uplink | 1 | | UL Information Transfer (Security mode complete) (Registration request) | | | | | |
| 2287 | 2022-04-11 10:24:22.928793000 | downlink | 1 | | Security Mode Command | | | | | |
| 2290 | 2022-04-11 10:24:22.939869000 | uplink | 1 | | Security Mode Complete | | | | | |
| 2331 | 2022-04-11 10:24:23.053802000 | downlink | 1 | | UE Capability Enquiry | | | | | |
| 2337 | 2022-04-11 10:24:23.066185000 | uplink | 1 | | UE Capability Information | | | | | |
| 2406 | 2022-04-11 10:24:23.208771000 | downlink | 1 | | DL Information Transfer (Registration accept) | | | | | |
| 2408 | 2022-04-11 10:24:23.211867000 | uplink | 1 | | UL Information Transfer (Registration complete) | | | | | |
| 2669 | 2022-04-11 10:24:24.212825000 | uplink | 1 | | UL Information Transfer (UL NAS transport) (PDU session establishment requ... | | | | | |
| 2703 | 2022-04-11 10:24:24.285803000 | downlink | 1 | | RRC Reconfiguration (DL NAS transport) (PDU session establishment accept)[... | | | | | |
| 2704 | 2022-04-11 10:24:24.292873000 | uplink | 1 | | RRC Reconfiguration Complete | | | | | |
| 2974 | 2022-04-11 10:24:25.318843000 | uplink | 1 | | Echo (ping) request id=0xc641, seq=0/0, ttl=10 (reply in 3024) | | | | | |
| 3024 | 2022-04-11 10:24:25.469799000 | downlink | 1 | | Echo (ping) reply id=0xc641, seq=0/0, ttl=64 (request in 2974) | | | | | |
| 3160 | 2022-04-11 10:24:26.315348000 | uplink | 1 | | Echo (ping) request id=0xc641, seq=1/256, ttl=10 (no response found!) | | | | | |
| 3390 | 2022-04-11 10:24:27.315314000 | uplink | 1 | | Echo (ping) request id=0xc641, seq=2/512, ttl=10 (reply in 3415) | | | | | |
| 3415 | 2022-04-11 10:24:27.379798000 | downlink | 1 | | Echo (ping) reply id=0xc641, seq=2/512, ttl=64 (request in 3390) | | | | | |
| 3649 | 2022-04-11 10:24:28.315339000 | uplink | 1 | | Echo (ping) request id=0xc641, seq=3/768, ttl=10 (reply in 3666) | | | | | |
| 3666 | 2022-04-11 10:24:28.339798000 | downlink | 1 | | Echo (ping) reply id=0xc641, seq=3/768, ttl=64 (request in 3649) | | | | | |
| 3910 | 2022-04-11 10:24:29.315339000 | uplink | 1 | | Echo (ping) request id=0xc641, seq=4/1024, ttl=10 (reply in 3927) | | | | | |
| 3927 | 2022-04-11 10:24:29.339799000 | downlink | 1 | | Echo (ping) reply id=0xc641, seq=4/1024, ttl=64 (request in 3910) | | | | | |
| 4172 | 2022-04-11 10:24:30.315316000 | uplink | 1 | | Echo (ping) request id=0xc641, seq=5/1280, ttl=10 (reply in 4188) | | | | | |
| 4188 | 2022-04-11 10:24:30.339801000 | downlink | 1 | | Echo (ping) reply id=0xc641, seq=5/1280, ttl=64 (request in 4172) | | | | | |
| 4432 | 2022-04-11 10:24:31.315343000 | uplink | 1 | | Echo (ping) request id=0xc641, seq=6/1536, ttl=10 (reply in 4449) | | | | | |
| 4449 | 2022-04-11 10:24:31.339802000 | downlink | 1 | | Echo (ping) reply id=0xc641, seq=6/1536, ttl=64 (request in 4432) | | | | | |
| 4693 | 2022-04-11 10:24:32.315349000 | uplink | 1 | | Echo (ping) request id=0xc641, seq=7/1792, ttl=10 (reply in 4710) | | | | | |
| 4710 | 2022-04-11 10:24:32.339802000 | downlink | 1 | | Echo (ping) reply id=0xc641, seq=7/1792, ttl=64 (request in 4693) | | | | | |
| 4956 | 2022-04-11 10:24:33.315313000 | uplink | 1 | | Echo (ping) request id=0xc641, seq=8/2048, ttl=10 (reply in 4971) | | | | | |
| 4971 | 2022-04-11 10:24:33.339801000 | downlink | 1 | | Echo (ping) reply id=0xc641, seq=8/2048, ttl=64 (request in 4956) | | | | | |
| 5215 | 2022-04-11 10:24:34.315333000 | uplink | 1 | | Echo (ping) request id=0xc641, seq=9/2304, ttl=10 (reply in 5232) | | | | | |
| 5232 | 2022-04-11 10:24:34.339803000 | downlink | 1 | | Echo (ping) reply id=0xc641, seq=9/2304, ttl=64 (request in 5215) | | | | | |
| 5481 | 2022-04-11 10:24:35.326869000 | uplink | 1 | | UL Information Transfer (Deregistration request (UE originating)) | | | | | |
| 5500 | 2022-04-11 10:24:35.351801000 | downlink | 1 | | DL Information Transfer (Deregistration accept (UE originating)) | | | | | |
| 5561 | 2022-04-11 10:24:35.553807000 | downlink | 1 | | RRC Release | | | | | |

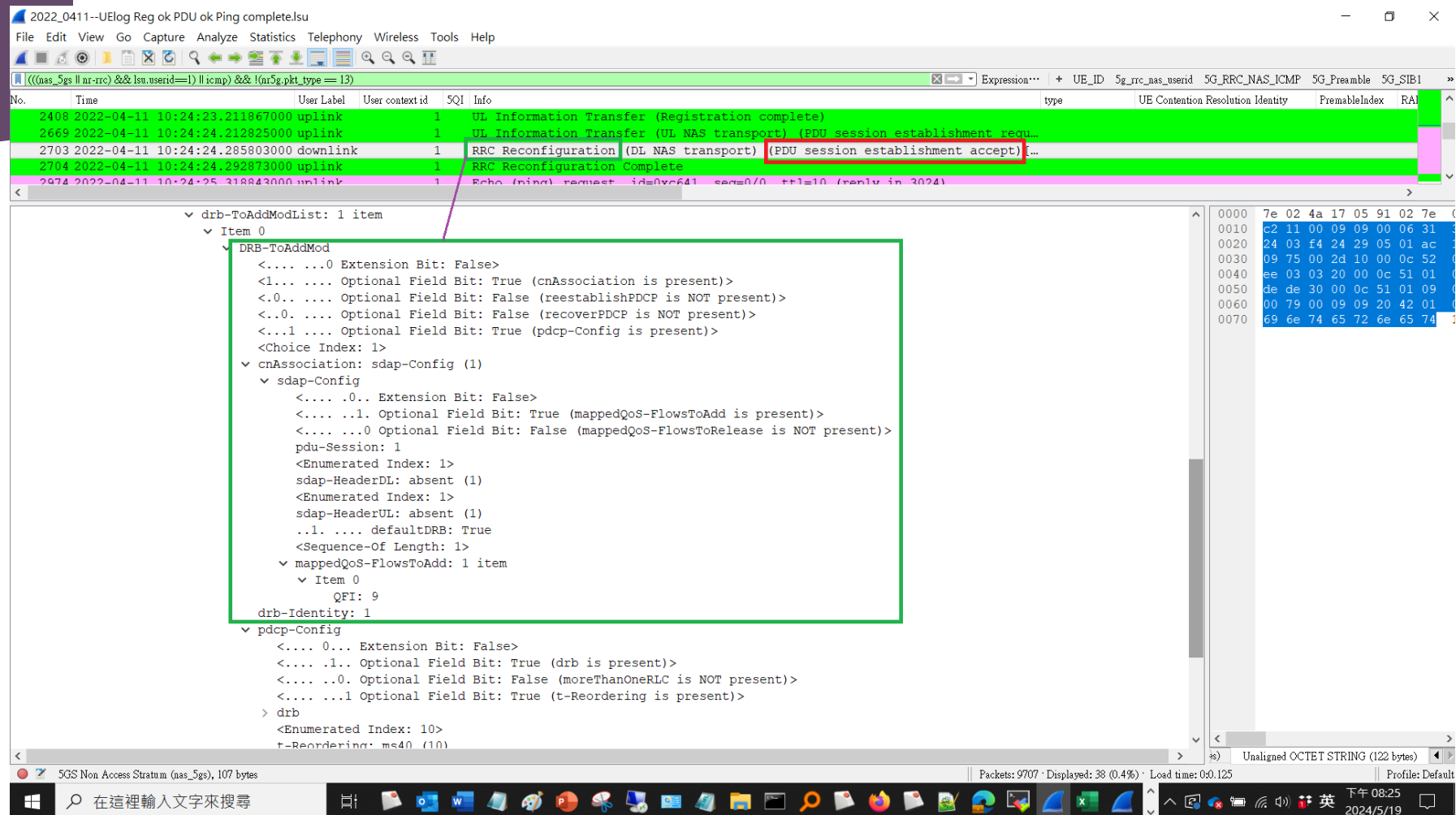
5GS Non Access Stratum (nas_5gs), 107 bytes

Packets: 9707 · Displayed: 38 (0.4%) · Load time: 0:0.125

Profile: Default

34

5G QoS Callflow – RRC Reconfiguration



5G QoS Callflow – translations for human--5

gNB to UE:

- mappedQoS-FlowtoAdd
- QoS Flow ID

5G QoS flows -- DL

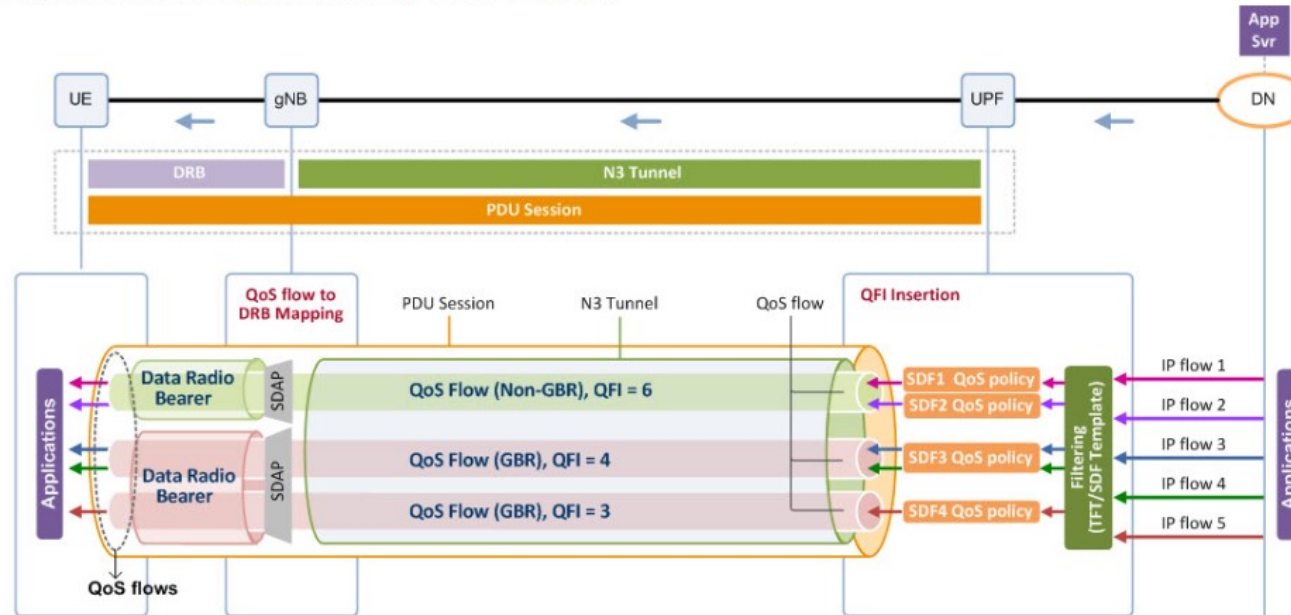
<https://netmanias.com/en/post/oneshot/14105/5g/5g-qos>

NETMANIAS ONE-SHOT

5G QoS - Downlink

February 7, 2019

The QoS differentiation within a PDU session



5QI : 5G QoS Identifier
 ARP : Allocation and Retention Priority
 GFBR : Guaranteed Flow Bit Rate
 MFBR : Maximum Flow Bit Rate
 PDB : Packet Delay Budget
 PER : Packet Error Rate
 QFI : QoS Flow Identifier
 RQA : Reflective QoS Attribute

| QoS Flow type | QoS Flow parameters |
|---------------|--------------------------|
| GBR flow | Non-GBR flow |
| | 5QI |
| | ARP |
| | RQA |
| | GFBR |
| | MFBR |
| | Notification Control |
| | Maximum Packet Loss Rate |

5QI
 Resource Type*
 Default Priority Level
 PDB
 PER
 Default Maximum Data Burst Volume
 Default Averaging Window
 * GBR, non-GBR or delay critical GBR

5QI Characteristics

- Example of GBR

5G QoS Identifier | 5QI Table

Following table mentions 5QI values and their corresponding QoS characteristics mapping.

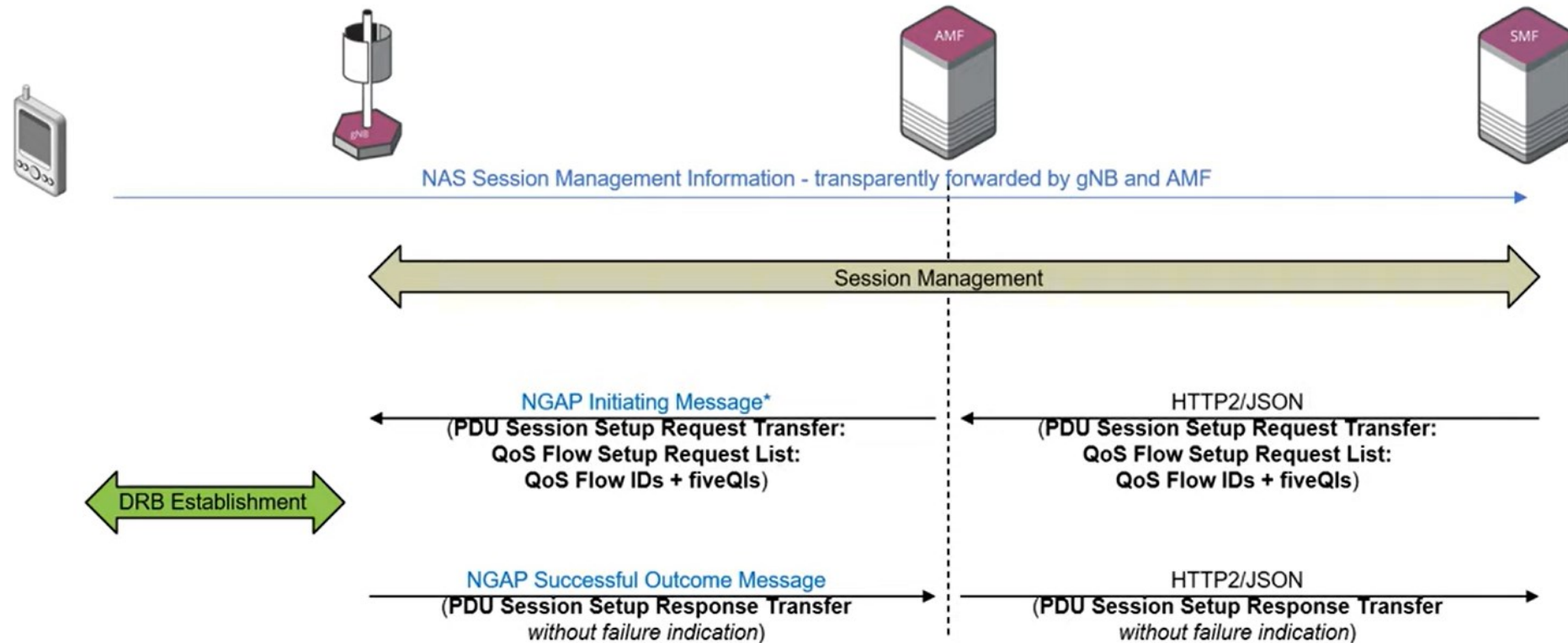
| 5QI Value | Resource Type | Default Priority Level | Packet Delay Budget | Packet Error Rate | Default Maximum Data Burst Volume | Default Averaging Window | Example Services |
|-----------|--------------------------------|------------------------|---------------------|-------------------|-----------------------------------|--------------------------|---|
| 1 | GBR (guaranteed flow bit rate) | 20 | 100 ms | 10^{-2} | N/A | 2000 ms | Conversational Voice |
| 2 | GBR | 40 | 150 ms | 10^{-3} | N/A | 2000 ms | Conversational Video (Live Streaming) |
| 3 | GBR | 30 | 50 ms | 10^{-3} | N/A | 2000 ms | Real time gaming, V2X messages |
| 4 | GBR | 50 | 300 ms | 10^{-6} | N/A | 2000 ms | Non-conversational Video (Buffered Streaming) |
| 65 | GBR | 7 | 75 ms | 10^{-2} | N/A | 2000 ms | Mission Critical user plane Push to Talk voice (e.g. MCPTT) |
| 66 | GBR | 20 | 100 ms | 10^{-2} | N/A | 2000 ms | Non-Mission Critical user plane Push to Talk Voice |
| 67 | GBR | 15 | 100 ms | 10^{-3} | N/A | 2000 ms | Mission Critical Video user plane |

5QI Characteristics

- Example of NGBR

| | | | | | | | |
|----|---------|----|------------|------------|------------|------------|--|
| 74 | GBR | 56 | 500 ms | 10^{-4} | N/A | 2000 ms | "Live" Uplink Streaming |
| 76 | GBR | 56 | 500 ms | 10^{-4} | N/A | 2000 ms | "Live" Uplink Streaming |
| 5 | Non-GBR | 10 | 100 ms | 10^{-6} | N/A | N/A | IMS Signalling |
| 6 | Non-GBR | 60 | 300 ms | 10^{-6} | N/A | N/A | Video (Buffered Streaming) TCP-based (e.g. www, e-mail, chat, FTP, p2p file sharing, progressive video etc.) |
| 7 | Non-GBR | 70 | 100 ms | 10^{-3} | N/A | N/A | Voice, Video (Live Streaming), Interactive Gaming |
| 8 | Non-GBR | 80 | 300 ms | 10^{-6} | N/A | N/A | Video (Buffered Streaming) TCP-based (e.g. www, e-mail, chat, FTP, p2p file sharing, progressive video etc.) |
| 9 | Non-GBR | 90 | -As above- | -As above- | -As above- | -As above- | -As above- |
| 69 | Non-GBR | 5 | 60 ms | 10^{-6} | N/A | N/A | Mission Critical Delay Sensitive Signalling (e.g. MC-PTT Signalling) |
| 70 | Non-GBR | 55 | 200 ms | 10^{-6} | N/A | N/A | Mission Critical Data (e.g. example services are the same as 5QI 6/8/9) |

QoS Flow Setup: gNB - AMF - SMF



* This can be:
NGAP Initial Context Setup Request
NGAP PDU Session Resource Setup Request
NGAP PDU Session Resource Modify Request
NGAP Handover Request (with different success/failure Response message)

5G QoS Flow/SDAP

QCI(4G) → 5QI(5G):

- * delay critical GBR ← additional GBR with Packet delay budget, Packet delay
(intelligent transport, electricity distribution, discrete automation)
- * Low latency eMBB (AR-augmented Reality)
- * High-speed trains

+ new 5G layer:
SDAP
(with Reflective QoS)

TS 23.501

Intelligent transport

reflective quality of service let's talk about it a

5G QoS Flow/SDAP

SDAP layer for:

- * User plane only
- * DL/UL QoS Flow mapping to DRB
- * Markering packets with QoS Flow ID (QFI)
- * Reflective QoS Flow to DRB for UL.
- * independent AS and NAS
- * maybe transparent

QoS Flows

SDAP

Radio Bearers

PDCP

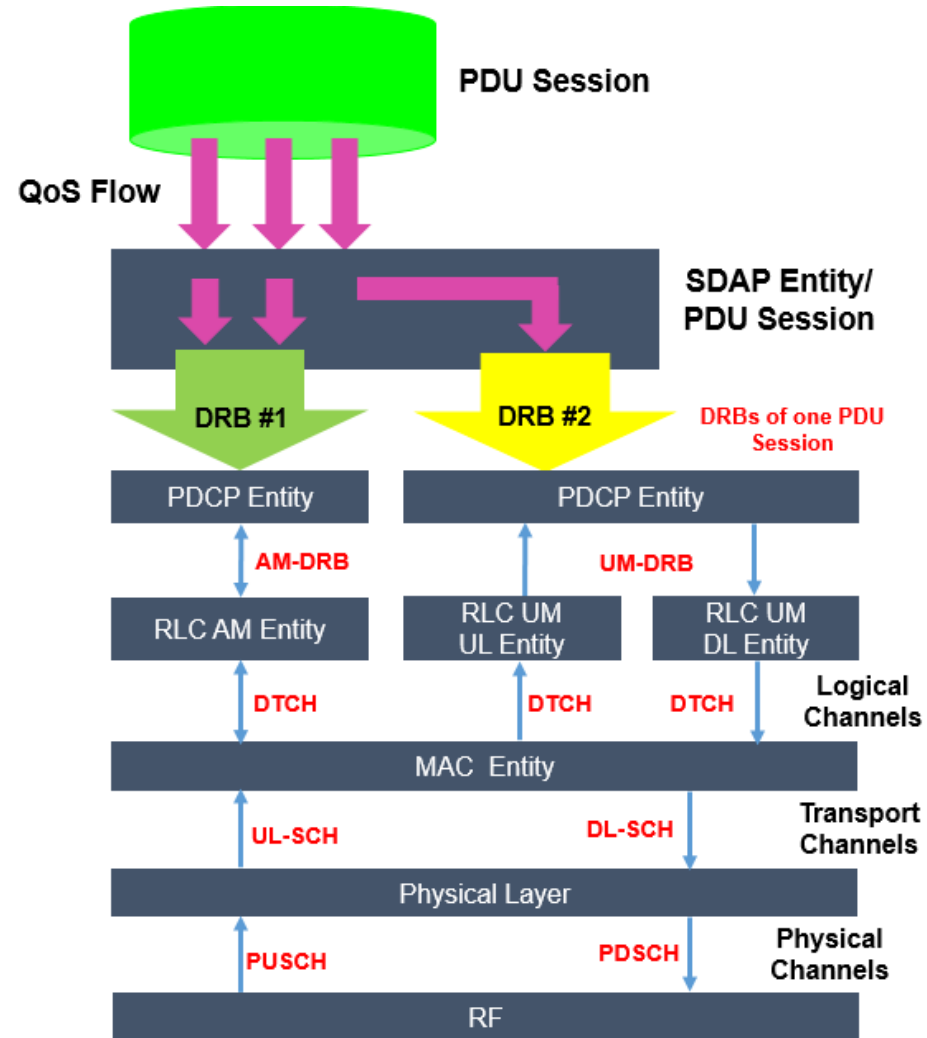
RLC

MAC

PHY

IoT Understanding

Important pages



5G QoS Callflow – Ping

2022_0411--5Gcsimlog Reg ok PDU ok Ping is less but ok, pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ngap||gtp

| No. | Time | Source Address | Destination Address | Protocol | 5QI | Info |
|------|----------------------------|----------------------------|-----------------------------|------------|-----|--|
| 2004 | 2022-04-11 18:01:47.850104 | 192.168.50.102, 172.16.0.1 | 192.168.50.101, 22.22.22.22 | GTP <ICMP> | | Echo (ping) request id=0xc641, seq=0/0, ttl=10 (reply in 2005) |

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 128
Identification: 0x388a (14474)
> 010. = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0x1bc7 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.50.102
Destination Address: 192.168.50.101

✓ User Datagram Protocol, Src Port: 2152, Dst Port: 2152
Source Port: 2152
Destination Port: 2152
Length: 108
Checksum: 0xc835 [unverified]
[Checksum Status: Unverified]
[Stream index: 3]
> [Timestamps]
UDP payload (100 bytes)

✓ GPRS Tunneling Protocol
> Flags: 0x34
Message Type: T-PDU (0xff)
Length: 92
TEID: 0x00000004 (4)
Next extension header type: PDU Session container (0x85)
✓ Extension header (PDU Session container)
Extension Header Length: 1
✓ PDU Session Container
0001 = PDU Type: UL PDU SESSION INFORMATION (1)
... 0000 = Spare: 0x0
00.. = Spare: 0x0
..00 1001 = QoS Flow Identifier (QFI): 9
Next extension header type: No more extension headers (0x00)

✓ Internet Protocol Version 4, Src: 172.16.0.1, Dst: 22.22.22.22
0100 = Version: 4

0000 00 0c 29 89 8a 28 00 00 fc 02 06 94 08 00 45 00 ..(.....E-
0010 00 80 38 8a 40 00 40 11 1b c7 c0 a8 32 66 c0 a8 ..8.@. @. ...2f..
0020 32 65 08 68 08 68 00 6c c8 35 34 ff 00 5c 00 00 2e h h l 54 ..\..
0030 00 04 00 00 00 85 01 10 09 00 45 00 00 54 00 00E..T..
0040 00 00 0a 01 d8 6c ac 10 00 01 16 16 16 16 08 001..
0050 a3 e2 c6 41 00 00 62 53 91 59 00 04 db 57 00 00 ...A..bS .Y...W..
0060 00 00 00 00 00 00 10 11 12 13 14 15 16 17 18 19
0070 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27 28 29! " # \$ % & ' ()
0080 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 36 37 *+,./01 234567

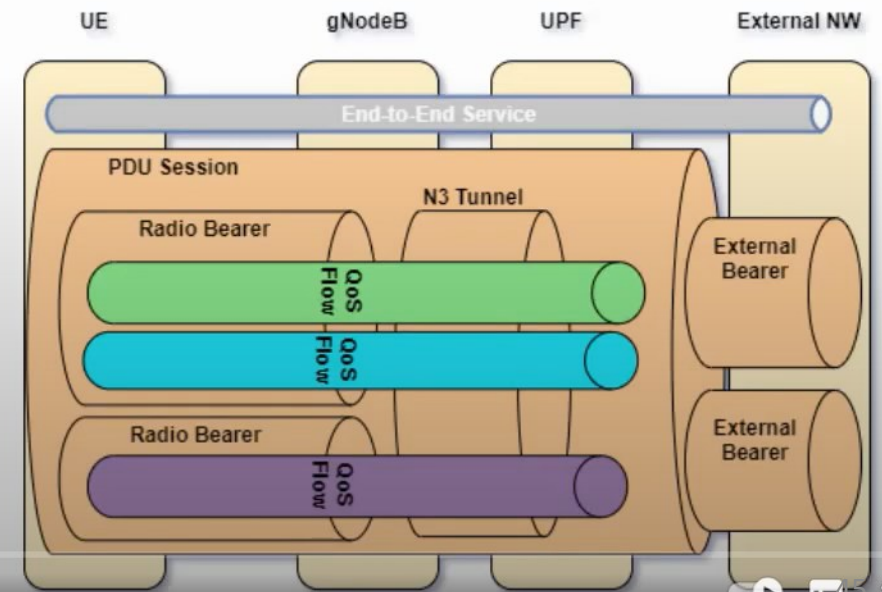
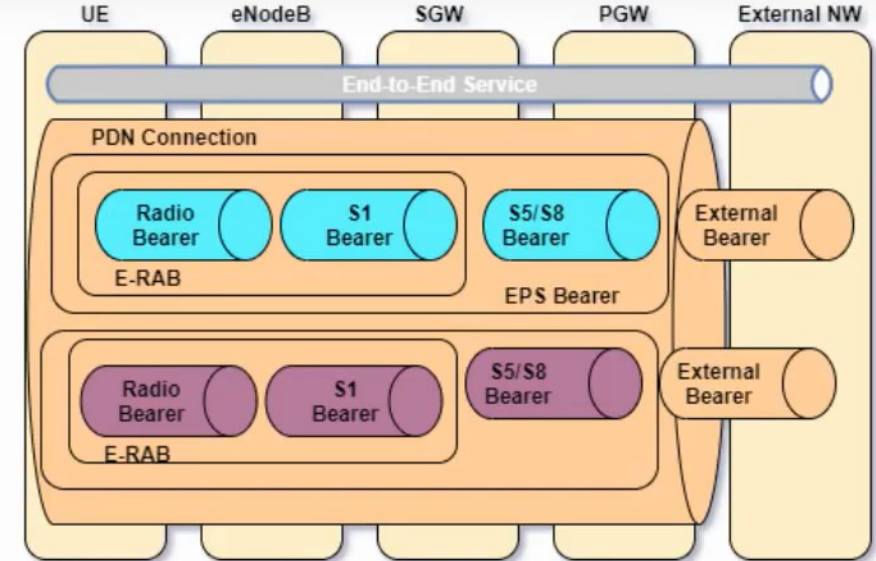
Packets: 2082 · Displayed: 47 (2.3%) Profile: Default

Text item (text)

在這裡輸入文字來搜尋

下午 08:42 2024/5/19

- In 5G QoS framework is based on QoS Flows which is the finest granularity of QoS differentiation.
- Each QoS Flow is identified by a QoS Flow ID.
- NG-RAN can map multiple QoS Flow IDs to one Radio Bearer or can have a 1:1 relationship. It depends on NG-RAN implementation.
- In 4G we had a 1:1 relationship between QCI and EPS bearers.
- 3GPP has standardized QoS Flow IDs like QCI for various services. More on this later.
- Also notice there is one N3 tunnel between eNB and UPF unlike multiple S5/S8 bearers in LTE.

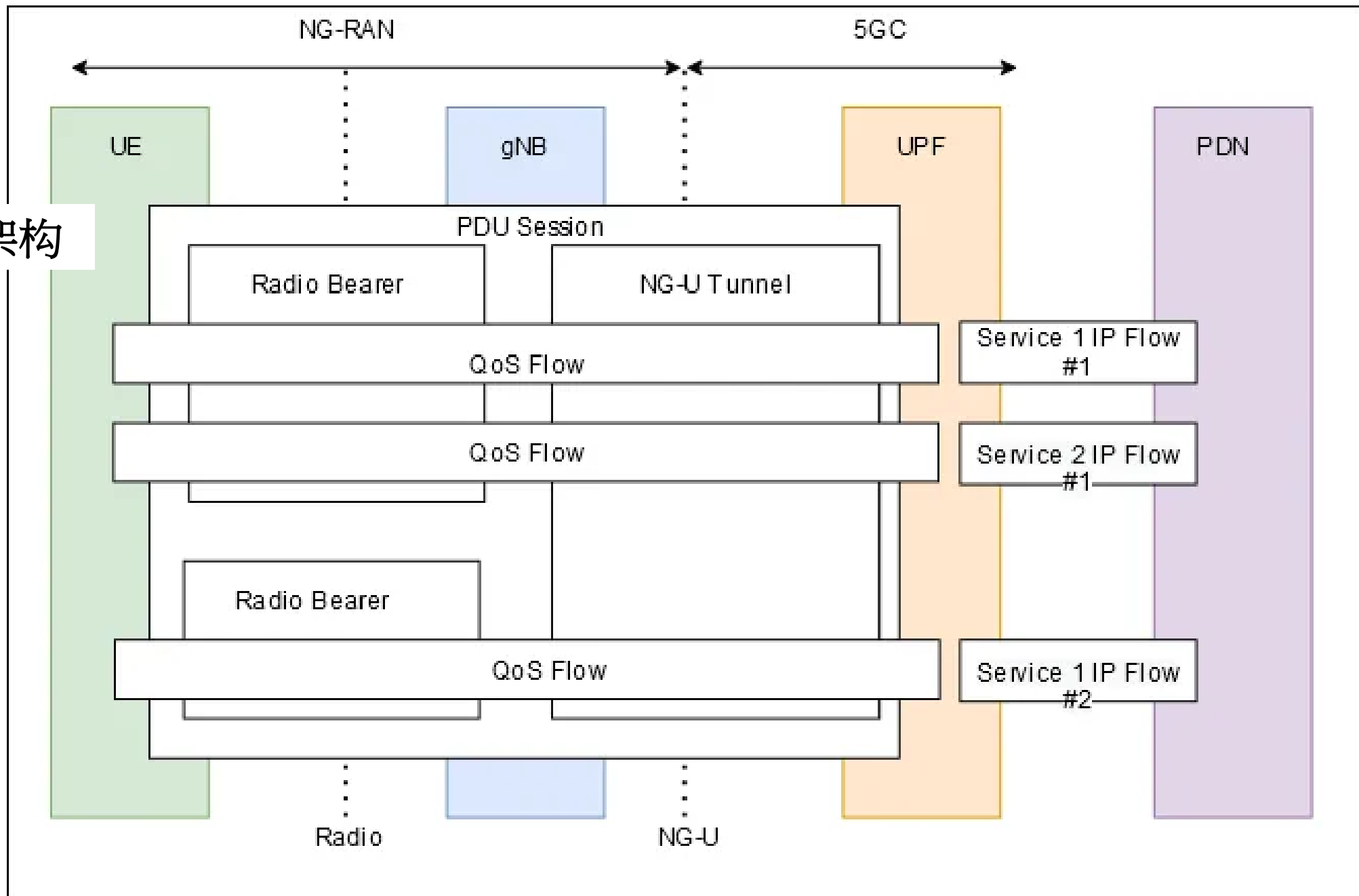


5G QoS vs 4G QoS

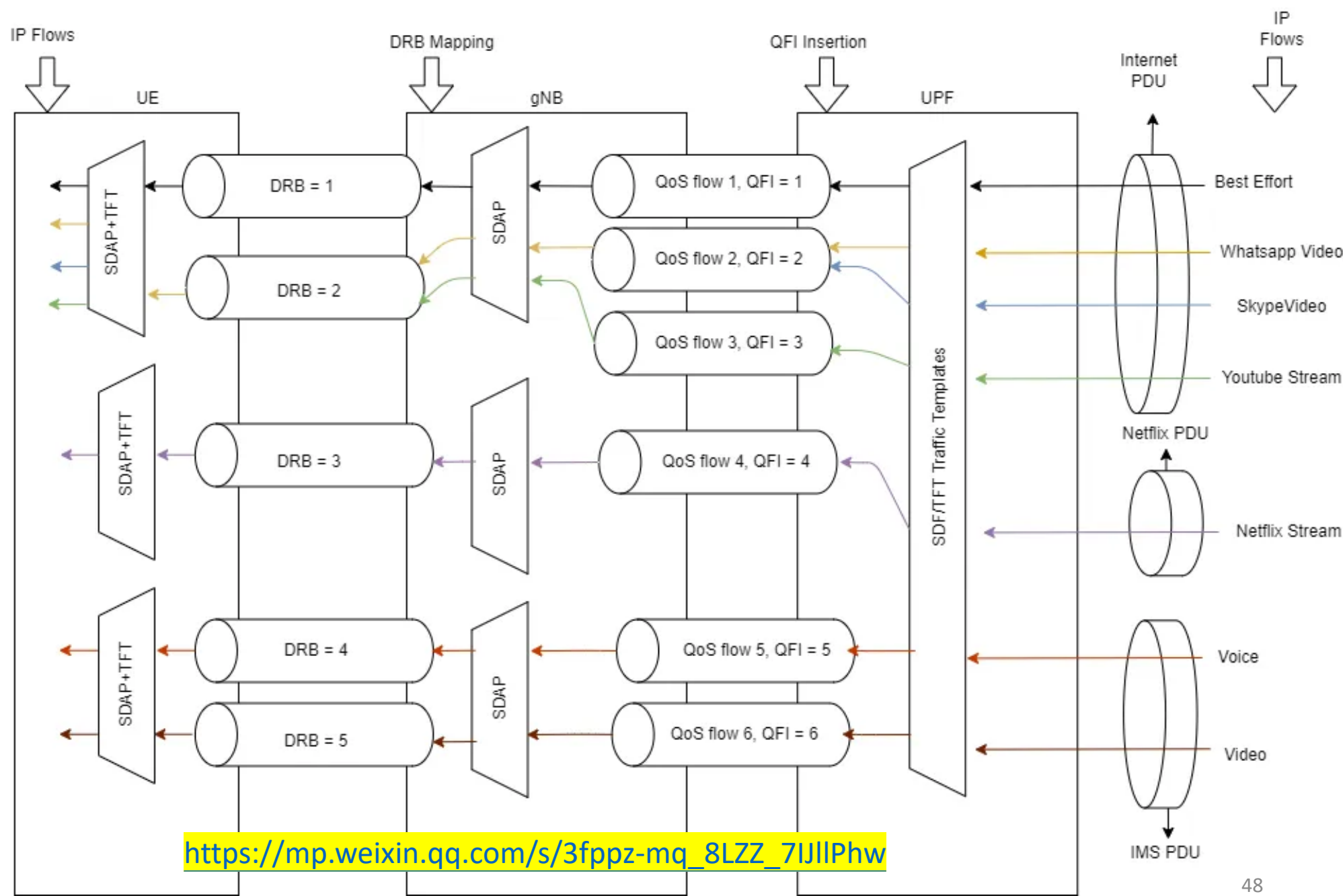
https://mp.weixin.qq.com/s/3fppz-mq_8LZZ_7IJlPhw

| Parameter | 5G | 4G |
|----------------|---------------------------|----------------------------|
| QoS Identifier | 5G QI (QoS Identifier) | QCI (QoS Class Identifier) |
| IP Data Flow | QoS Flow | EPC Bearer |
| Flow/Bearer ID | QFI (QoS Flow Identifier) | EBI |
| Reflective QoS | RQI (Reflective QoS ID) | NA |
| Data Session | PDU Session | PDN Connection |

5G QoS架构

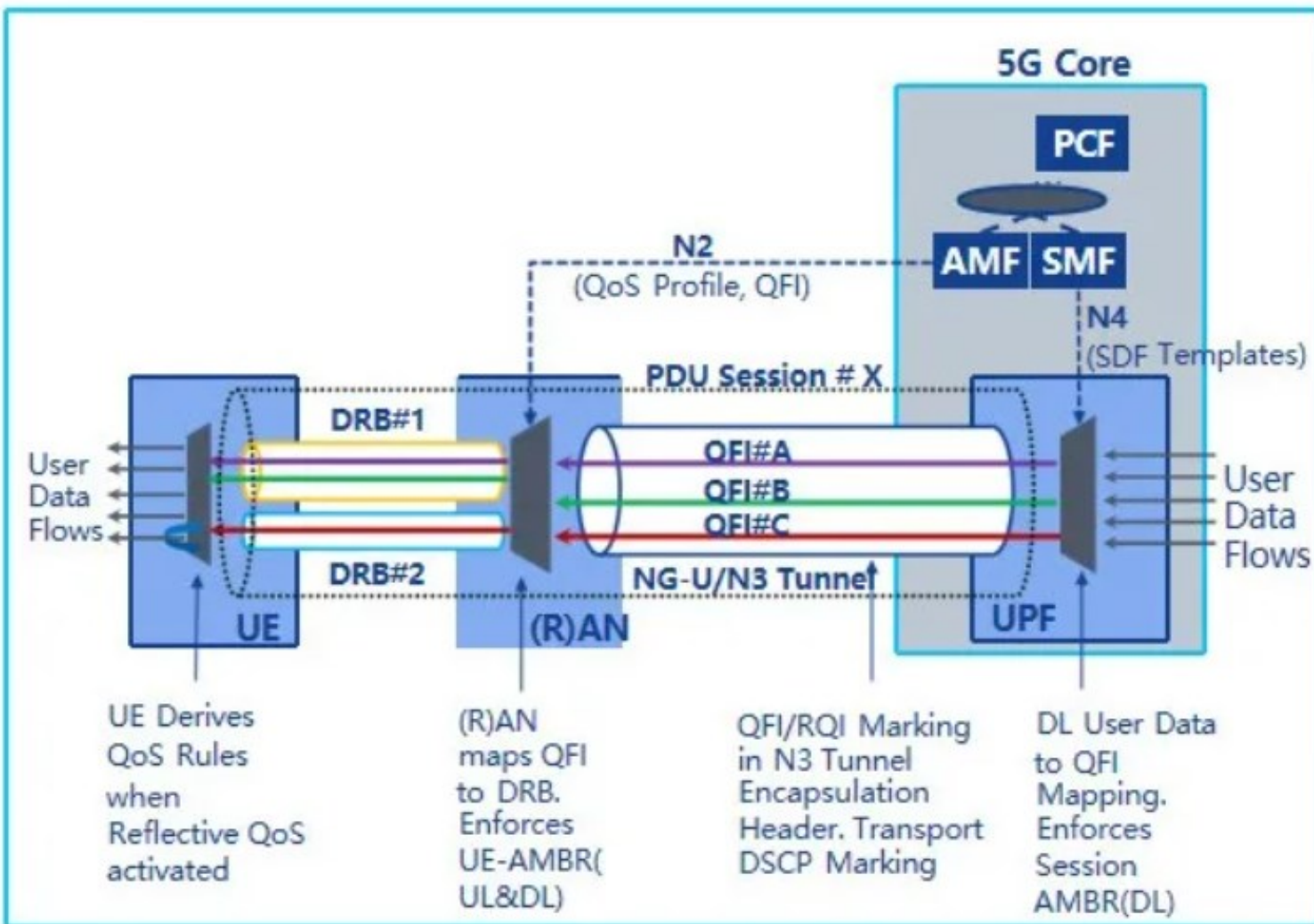


5G QoS示例



https://mp.weixin.qq.com/s/3fppz-mq_8LZZ_7IJlPhw

下行QoS流处理



UPF将服务数据流 (SDF) 映射到QoS流

UPF执行Session AMBR并且还可以对PDU计数以支持计费

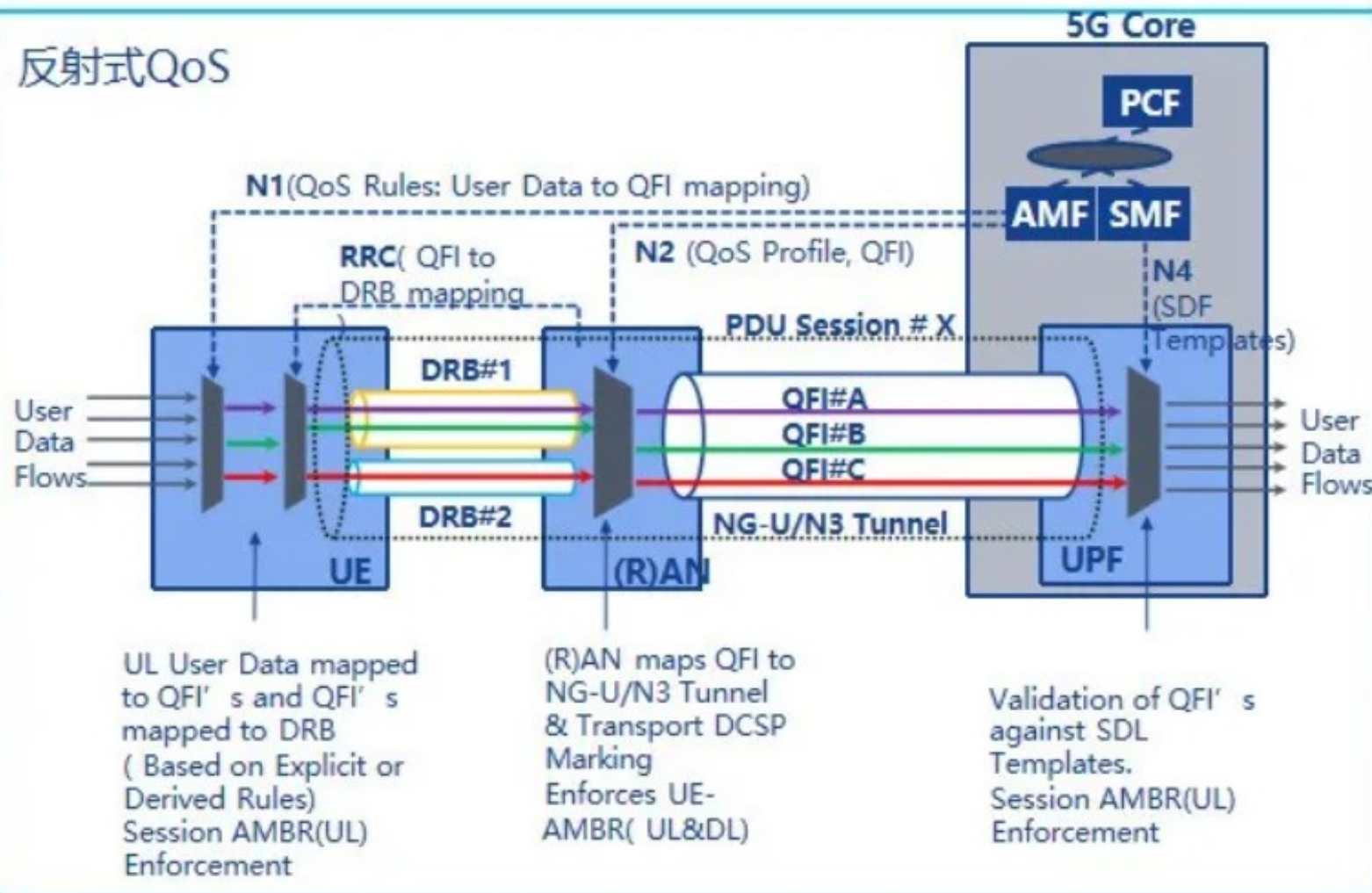
UPF在5GC和(R)AN之间的单个隧道中传输PDU会话的数据包，UPF封装报头中的用户平面标记（包括用于非3GPP接入的5QI）。另外，UPF可以在封装报头中包括对反射式QoS激活的指示。

(R)AN根据QFI和相关联的5G QoS特性和参数将来自QoS流的PDU映射到特定接入的资源。

如果应用反射QoS，则UE创建新的派生QoS规则。派生QoS规则中的分组过滤器来自下行分组包（即，报头），并且派生的QoS规则的用户面标记来自于下行分组包的用户面标记。

上行QoS流处理

反射式QoS



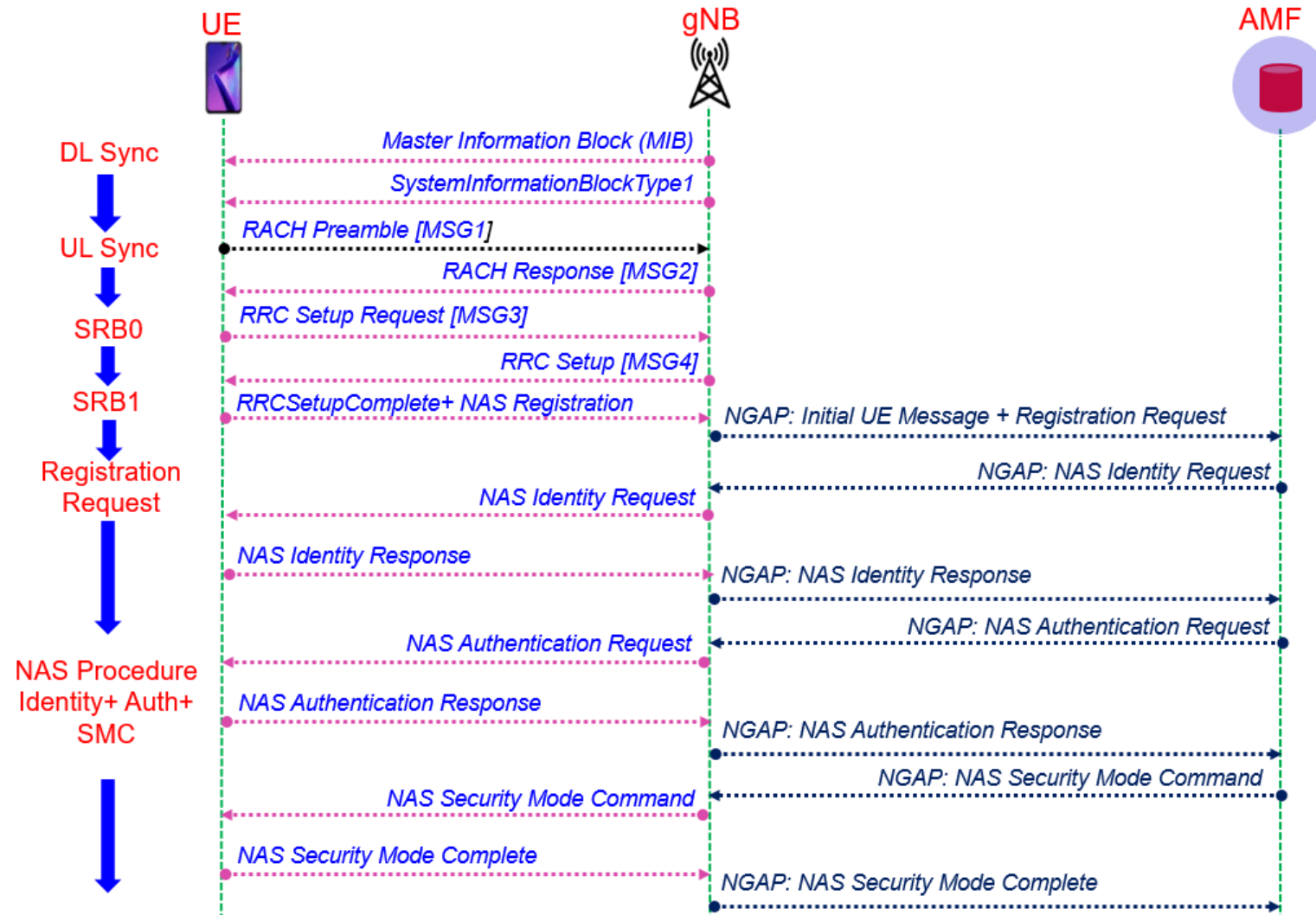
UE使用存储的QoS规则来确定SDF与QoS流之间的映射。UE基于RAN提供的映射,使用与QoS流相对应的接入特定资源来传输上行PDU。

(R)AN经由N3隧道向UPF传输PDU。在从(R)AN向CN传递上行分组包时,(R)AN可确定QFI值(该值包含在UL PDU的封装报头中),并选择N3隧道。

UPF验证UL PDU中的QFI是否与提供给UE或由UE隐式派生(例如,在采用反射式QoS的情况下)的QoS规则相一致。

UPF执行session AMBR并对PDU计数以进行计费

5G E2E Call flow -- SYNC to NAS SMC



5G E2E Call flow -- Registration to PDUsession



Information for Quiz of 5G QoS lesson

- 5 questions for Acronyms of QoS parameters
- 5 questions for definitions/explanations of QoS parameters/QoS procedures
- Two questions for bonus

Thank You

